An Enclosed Mode Is a Gauge Choice: Topology Relative to Reach in Certified Code World Models

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Abstract

A code world model accepted by a sampling gate can be exactly right on everything the gate can see and arbitrarily wrong beyond it. We characterize what a certified model can know — and what its errors can cost — when the omitted structure is a topologically nontrivial region: an annular freeze mode enclosing an unreachable interior. The *gate quotient* makes the question precise: acceptance-with-certainty determines the model exactly up to the reachable query set, so everything beyond reach is gauge. On a minimal ring instrument we prove the extreme case — a wrong-topology filled-disc artifact is unfalsifiable by *any* sampling gate and bitwise harmless at play — and measure, with real LLM synthesis across three model families, how a single knob (an angular channel of width γ) walks the same artifact through three regimes: unfalsifiable-and-harmless, falsifiable-and-costly, and instantly falsified. Two results organize the empirics. First, **danger is topology relative to reach, not topology**: opening a channel the planner can use collapses the blind model's exploitation (`play_cost` $1.09 \to \sim 0$ over a knee at $\gamma \approx 0.1$), while a hidden channel with the *same* first Betti number keeps it at full strength (1.12). Second, **repair is parameter-bound and sensor-bound**: from outside evidence no family recovers the region (superstitious point fits); from inside, models pose the right topology but cannot pin its parameters to the gate's precision — and the persistent-homology summary that guides them has a measurable resolution limit — geometric, not budgetary — and the topology the loop's artifacts pose tracks that summary's wrong $\hat{\beta}_1$, not the truth. Planner-side mitigation obeys the same geometry: point distrust-fences, a zero-dimensional cover, fail against the one-dimensional closed boundary at their calibrated radius — while a fence matched to the boundary's dimension and *persisted* across episodes collapses the exploitation to a two-lesson transient (play_cost $0.999 \to 0.058$, truth-equal returns from the second episode on). Against an *invented* mode every fence fires constantly and recovers nothing — and the dual certificate (freedom points that locally un-freeze imagination where the model was refuted as too pessimistic) collapses that failure symmetrically ($1.769 \to 0.029$): the two wrongnesses need opposite defenses, each priced by its failure's lie rate. In n dimensions the enclosing shell makes the identifiability event near-certain ($r(n) \to 0$) while the danger stays fully exploitable — rarity and reachability are independent knobs of the danger law.

1 Introduction

In the Code World Model (CWM) paradigm, a language model synthesizes an executable world model that a classical planner searches, and the model is accepted when it reproduces sampled transitions. Two companion papers established that this gate certifies the wrong thing, following a quantitative law, danger = play_cost $\times (1 - \text{rarity})^N$: in discrete games (Aguilar Martín, 2026b), and in continuous hybrid systems, where an omitted mode is a rare rule and the danger collapses to pure identifiability (Aguilar Martín, 2026a). The continuous paper ended on a sharpened negative: repair-from-data is geometry-dependent, and its two ablations located the mechanism in a *template* *prior* — the synthesizer selects low-descriptive-complexity region forms (1D thresholds, radial balls) instead of inducing the boundary, flattening curves and curving flats alike. Curvature is not the axis. This paper takes up the axis that survived: **topology** — and shows that the right formulation is not the topology of the mode but its topology *relative to what the certifying process and the planner can reach*.

Why an enclosed mode? Because it is the minimal setting where the certification question and the topology question are the same question. An annular freeze band — a fence with an inside — is the simplest region whose defining feature, the hole, is invisible from one side: every trajectory that approaches from outside stops at the outer wall, and whether anything is behind it is, in a sense this paper makes exact, *not a property of the data at all*. Fences, containment shells, and geofenced no-go regions are the practical shape of safety-critical hybrid modes; a certification pipeline that cannot distinguish a fenced void from a fenced hazard — or a fence from a filled wall — is certifying less than it appears to. The question is what the gate's acceptance actually pins down, and the answer turns out to be geometric: everything up to reach, nothing beyond it, with the boundary between the two movable by knobs that never touch the mode's own physics.

The instrument is deliberately minimal (Section 2): a 2D thrust-and-drag plant with an annular freeze band (inner radius 3.5, outer 5.0) wrapping a high-reward "phantom" lode, with three pre-registered knobs — an angular channel of width γ (closed ring at $\gamma = 0$), the channel's orientation (facing the start or hidden behind the lode), and the start side (outside or inside the hole). Everything the paper claims follows from what these knobs do to the *reachable query set* $\mathcal{R}$.

Contributions.

1. **The gate quotient and the unfalsifiable-and-harmless theorem** (Section 3). Acceptance-with-certainty determines a model exactly up to μ_{query}-null modification on $\mathcal{R}$; beyond $\mathcal{R}$ is gauge (Proposition 1). On the closed ring, the wrong-topology filled-disc artifact is unfalsifiable by any sampling gate *and* bitwise planner-equivalent to the truth (Proposition 2, confirmed episode-for-episode). Monotonicity, continuity, and positivity of the rarity curves in γ round out the provable core (Section 3.1, stated and proved, one positivity witness machine-checked); the two monotonicity statements hold up to an explicit *funnel defect* (Corollary 2) that we bound by measurement, 17× below the effect — unconditionally they are out of reach, and a measured certificate shows why no pathwise proof can exist (Remark 5).

2. **The three-regime mechanism, without LLMs** (Section 4). One artifact walks through unfalsifiable-and-harmless ($\gamma = 0$), falsifiable-and-costly (facing $\gamma > 0$), and instantly-falsified (inside start) as the knobs move. Two readings matter later: an *invented* mode is exploited *below random* from inside (play_cost 1.77 — the dual of phantom-free-space exploitation), and the hidden channel is observationally *identical* to the closed ring — same β_1 change, no observable consequence.

3. **LLM synthesis on the closed ring, three families** (Section 5). The identifiability event is family-independent (GPT-5.x, Qwen, Claude: every mode-absent sample yields a certified, blind, exploited model at play_cost ≈ 1.12). None of the three tested families repairs the ring from outside evidence — a behavioral audit shows the gate-passing "repairs" are measure-zero point fits. From inside, models *pose* the right topology but repair is **parameter-bound**: the single gate-certified recovery (1/20) used the gauge-free disc-complement form whose one parameter is the true round radius anchored in the reward specification — and the strongest cross-family repairer (Claude, 3/3) used exactly the same form.

4. **The open-ring arm: danger is topology relative to reach** (Section 6). Pre-registered sweep over γ: the blind model's exploitation collapses over a knee at $\gamma \approx 0.1$ (channel arc-width comparable to the planner's step) when the channel faces the start, and *persists at full strength* (1.116) at the same γ when the channel is hidden — identical β_1, opposite danger. Gate-pass from inside stays $\approx$0 at every γ, and the two certified passes at wide γ are *certified-wrong*: mode-blind on the probes, harmless only because the open ring no longer obstructs.

5. **Mitigation obeys a covering law — and the defense that works pays it once** (Section 7). The companion paper's distrust-fence defense, verbatim, does nothing on the ring at its calibrated radius: point fences are a zero-dimensional cover of a one-dimensional boundary, and the planner re-crosses through unfenced imagined corridors (cost = boundary measure over fence radius). Matching the fence's dimension to the boundary's (violations linked into tangentially-extended segments) *and persisting it across episodes* collapses the exploitation to a two-lesson transient: play_cost 0.999 $\to$ 0.058, with returns equal to the truth planner's from episode 2 on — the fence engineers the blind model back into the truth's gate-equivalence class on the operative side. Against the *invented* mode every distrust variant fires constantly and changes nothing — and the *dual* defense (freedom patching: un-freeze imagination, via the pinned integrator, where the model was refuted as too pessimistic) collapses that failure at once (1.769 $\to$ 0.029, near-truth returns from episode 1). The two wrongnesses need opposite defenses, and each defense's cost is set by its failure's lie rate: false obstructions refute themselves at every step; false freedoms only at the rare boundary.

6. **The evidence sensor has finite resolution — and the loop follows its report** (Section 8). The pre-registered persistent-homology summary reports $\hat{\beta}_1 = 1$ for every channel narrower than $\sim$2 arc-units (flip at $\gamma \approx 1.8$) even though the true $\beta_1 = 0$ for all $\gamma > 0$; the posed artifact topology tracks the *guidance*, not the truth. An honest, finite-resolution sensor causes wrong-topology gate-certified models. On the n-dimensional shell, inside-start evidence recovers β_{n-1} up to $n = 5$ where outside evidence recovers it at *no* n — recovery is start-governed, and the Niyogi–Smale–Weinberger density bound makes the outside failure computable in advance.

7. **Dimension as the rarity knob; the two axes are independent** (Section 9). On ShellField-n the contact rarity collapses geometrically (measured factor 0.411 per dimension; the exponential rate is proved under the isotropic action interface, and an explicit $4h/(n\kappa^2)$ bound for the instrument's box thrust; contacts 1/600 at $n = 4$, censored zeros at $n = 5, 6$), making mis-synthesis near-certain, while a competent vector planner is exploited at play_cost ≈ 1.0 at *every* $n \leq 6$ — rarity (synthesis axis) and reachability (play axis) are independent knobs, and a high-n enclosed mode maxes out both.

8. **A behavioral audit methodology** (used throughout): probe each artifact's `step()` on a state grid, classify the shape of its deviation-from-integrator set. It reproduces the companion paper's hand inspection, reclassifies the inside-start failures, and separates "posed the wrong structure" from "posed the right structure with unidentifiable parameters."

The practitioner's summary is one sentence long: *a sampling gate certifies the reachable restriction of the model and nothing else* — so the safety question is never "is the model right?" but "does the place where it is wrong intersect the operative reach of a competent planner?" Everything in this paper — the harmless wrong topology, the danger knee, the hidden-channel persistence, the parameter-bound repairs, the sensor's resolution limit — is that one sentence measured from six directions.

2 The instrument: a mode with an inside

`RingField2D` is the companion paper's plant with one structural change. The state is (x, y, v_x, v_y); a scalar action $a \in [-a_{\max}, a_{\max}]$ maps to a thrust heading $\phi = \pi a / a_{\max}$; velocities integrate with gain 3.0 and drag 0.3 at $dt = 0.1$ (the same semi-implicit integrator family as PatchField2D, so every planner and every synthesis-protocol constant carries over unchanged). The reward is two radial sigmoid lodes: a small *real* one (amplitude 0.3) behind the start at $(-6, 0)$, and a large *phantom* one (amplitude 1.0) at $c = (12, 0)$. The structural change is the mode: an **annular freeze band** $A = \{\, r_{\text{in}} \leq \|p - c\| \leq r_{\text{out}} \,\}$ with $r_{\text{in}} = 3.5$ and $r_{\text{out}} = 5.0$, wrapping the phantom lode. A step whose next position lands in A freezes at the *previous* position with zero velocity — the same inelastic-stop semantics as the companion paper's patches, but now the region has an inside: the ring separates the plane, and its defining feature ($\beta_1 = 1$, the hole) is exactly the part no outside observer ever touches.

Three pre-registered knobs deform the relationship between the mode and what is reachable, without touching the mode's local physics:

- **Channel width** γ (radians): an angular gap cut from the ring. $\gamma = 0$ is the closed ring ($\beta_1 = 1$); any $\gamma > 0$ is an open C-shape ($\beta_1 = 0$, contractible).
- **Channel orientation**: centered facing the start (`gap_center` $= \pi$) or hidden on the far side (`gap_center` $= 0$). Both have identical topology at equal γ; only the channel's position relative to the start and the planner's optimal path differs.
- **Start side**: outside the ring (the default; the interior is then unreachable at $\gamma = 0$, Lemma 1) or inside the hole (the interior evidence regime).

The defaults were frozen once at calibration and never tuned per cell: episode horizon $h = 80$ steps; thickness $w = 1.5$ exceeds the maximal per-step displacement $\Delta \leq 1.0$ (Section 3), so the no-jump-over lemma holds with margin for real and imagined rollouts alike. The outside contact rarity at the defaults is $r = 0.0312$ per rollout (937/30,000 calibration rollouts; the mechanism grid's own 400-rollout realization reads $18/400 = 0.045$, statistically compatible). The synthesis protocol is the companion paper's, verbatim: $N = 40$ uniform-random rollouts as evidence and gate, the integrator pinned in the contract, exactness tolerance $\varepsilon = 10^{-9}$, at most 5 refine iterations, and `play_cost` the paired-seed normalized regret of planning on the artifact and executing in the truth. Section 9 generalizes the instrument to `ShellField-`n — the same two-lode geometry with a spherical shell S^{n-1} in the first two coordinates and a thrust-vector action $\vec{a} \in [-1, 1]^n$ (norm-capped) — so that the ambient dimension n becomes a fourth knob.

3 Theory: the gate quotient

Setup. Deterministic dynamics $f : S \times A \to S$ on $S \subseteq \mathbb{R}^d$, initial distribution μ_0, episode horizon h, and a *gate policy* ρ selecting actions (uniform-random in our gates, though nothing below needs that). The **reachable query set** $\mathcal{R}(f, \mu_0, \rho, h)$ is the set of pairs (s, a) at which some length-$\leq h$ trajectory under (f, μ_0, ρ) queries f with positive probability — taken in continuous spaces as the support of the induced occupation measure. A *sampling gate* of any size N draws trajectories under

(f, μ_0, ρ) and accepts a candidate $\hat{f}$ iff $\hat{f}$ reproduces every queried transition (within any tolerance, including exactly).

Proposition 1 (gate quotient)**.** *Let $\mathcal{R}(f, \mu_0, \rho, h)$ be the reachable query set of the truth f under the gate policy ρ, and let $\hat{f}$ be any model with $\hat{f}|_{\mathcal{R}} = f|_{\mathcal{R}}$. Then for every N the sampling gate accepts $\hat{f}$ with probability 1, and the trajectory law of any policy whose queries stay in $\mathcal{R}$ is identical under f and $\hat{f}$. Conversely, let μ_{query} be the gate's query-occupation measure (the expected number of queries per rollout landing in each measurable set of $S \times A$) and let D be the set where $\hat{f}$ disagrees with f beyond the gate's tolerance. Acceptance with probability 1 at every N forces $\mu_{\text{query}}(D) = 0$: the model may differ from f only on a μ_{query}-null subset of $\mathcal{R}$ and on the gauge complement $\mathcal{G} = (S \times A) \setminus \mathcal{R}$, which no sample drawn under ρ can falsify. In particular the extension class $E(f) = \{\hat{f} : \hat{f}|_{\mathcal{R}} = f|_{\mathcal{R}}\}$ is accepted with certainty, and is the full accepted class up to μ_{query}-null sets.*

Proof. By induction on the step index, a trajectory under (f, μ_0, π) whose queries lie in $\mathcal{R}$ is a trajectory under $(\hat{f}, \mu_0, \pi)$ with the same realizations: the state after step t is a function of μ_0, the action sequence, and the queried values of f, which agree with $\hat{f}$'s on $\mathcal{R}$. Gate draws use $\pi = \rho$, whose queries lie in $\mathcal{R}$ by definition (up to a null set), so acceptance statistics coincide; $f \in E(f)$ is accepted almost surely, hence so is every member. For the converse, let p be the probability that a single gate rollout queries D at least once. Rollouts are i.i.d., so the size-N gate accepts $\hat{f}$ with probability $(1-p)^N$, and acceptance with certainty at every N forces $p = 0$. Finally $p = 0$ iff $\mu_{\text{query}}(D) = 0$: the per-rollout count of D-queries is a nonnegative random variable, so its expectation vanishes iff the count is almost surely zero. □

Remark 1 (relation to the companion identifiability proposition). The companion paper's identifiability result conditions on the finite-sample event "the mode region was missed," of probability $(1 - r)^N \to 0$ when $r > 0$. Proposition 1 is its structural, N-independent limit: on $\mathcal{G}$ the miss probability is 1 for *every* N. The companion's prior caveat — a prior or the specification could still supply the mode — is exactly the statement that $E(f)$ is not a singleton.

Lemma 1 (metric crossing)**.** *If positions move in steps of norm at most Δ and the annulus $A = \{r_{\text{in}} \le \|p-c\| \le r_{\text{out}}\}$ has thickness $w = r_{\text{out}} - r_{\text{in}} > \Delta$, then any trajectory from outside to the open interior visits A at an intermediate step. Consequently, at $\gamma = 0$ with freeze-on-entry dynamics the open interior is reach-null: the inner disc $\times$ all actions lies in $\mathcal{G}$.*

Proof. $g(t) = \|p_t - c\|$ changes by at most Δ per step (distance-to-center is 1-Lipschitz). Let t^* be the first index with $g(t^*) < r_{\text{out}}$. If $g(t^*) < r_{\text{in}}$ then $g(t^*-1) \ge r_{\text{out}}$ forces a step longer than $w > \Delta$, a contradiction; so $p_{t^*} \in A$. Under the freeze semantics the first landing in A is replaced by the previous (outside) position with zero velocity, so g never drops below r_{in}; induct. □

At the frozen defaults the hypothesis holds with margin: speed obeys $\|v_{t+1}\| \le (1 - \text{drag}\, dt)\|v_t\| + \text{gain}\, dt$, so $\|v\| \le \text{gain}/\text{drag} = 10$ and $\Delta \le 1.0 < w = 1.5$ — for real *and* imagined rollouts, since planners use the same integrator and action clamp. Measured: 200 calibration rollouts visit zero interior states while reaching the ring itself.

Corollary 1 (evidence equivalence of disc and annulus from outside)**.** *From outside starts no landing ever falls at $d < r_{\text{in}}$, so the disc mode $\|p - c\| \le r_{\text{out}}$ and the annulus mode fire on exactly the same steps of every realization: the contact processes — and hence* all *evidence any gate, repair loop, or topological summary extracts — are pathwise identical. (Measured as an exact row-for-row equality of the two contact clouds at every sample size.) The companion paper's dimensional* *reduction is* rational given the evidence*: outside data cannot even pose the disc-versus-annulus question.*

Remark 2 (metric, not topological — where algebraic topology is earned)*.* Lemma 1 uses only that distance-to-center is 1-Lipschitz; it works verbatim for round shells S^{n-1} in $\mathbb{R}^n$ and needs no homology. Genuine algebraic topology is *earned* exactly where this proof dies: non-round separators (Jordan–Brouwer to even define "inside") and non-separating modes (winding and linking obstructions — for the solid torus this is carried out in Section 9.1: the linking dichotomy). The round ring deliberately sits on the metric side of that boundary; the topology enters through what the gauge region and its boundary class organize, not through the crossing proof.

Remark 3 (machine-checked)*.* The paper's deterministic results are formalized in Lean 4 over mathlib (`formal/`, library `Paper3Ring`; no `sorry`), each at the level its proof actually operates: Lemma 1 in both halves over an arbitrary pseudometric space, with the constants paragraph above ($\|v\| \leq$ gain/drag through freeze events, $\Delta = 1.0 < w = 1.5$ by `norm_num`); Corollary 1 as a pathwise identity of trajectories and contact processes; the gate quotient (Proposition 1) at realization level; Proposition 2 composed with the planner/environment loop — unfalsifiability over arbitrary closed-loop policies, and play cost exactly 0 for any deterministic planner acting on imagined rollouts; the firing coupling of Proposition 3 and the direct-entry transport of Proposition 5, with Corollary 2's defect algebra checked from its named hypotheses; the fence and patch sufficiency theorems (Propositions 8 and 9) from their coverage/dominance hypotheses; Proposition 6 closed up to the start box's positive probability (the 1-D tube reduction, the in-horizon window, and channel membership via Jordan's inequality); and every measure step of Proposition 4's modulus end to end — the landing law's exact uniformity, the circle–strip bound with rotation invariance as a measure statement, measurability of the freeze trajectory in the action sequence, and the h-step slice/union composition — leaving only each application's coordinate computation. The item-by-item map, the modelling conventions, and the exact boundary of what remains measured are recorded in the repository's formalization ledger.

Proposition 2 (wrong topology: unfalsifiable and harmless at $\gamma = 0$)**.** *Let f be the closed-ring dynamics and $\hat{f}_{\text{fill}}$ the filled-disc model (freeze on the whole disc $\|p-c\| \leq r_{\text{out}}$). Then (i) $\hat{f}_{\text{fill}} \in E(f)$: every sampling gate accepts it with probability 1; and (ii) any deterministic planner whose imagined rollouts start at real (outside) states produces* identical *real trajectories under $\hat{f}_{\text{fill}}$ and under f —* play_cost $= 0$ *exactly.*

Proof. (i) is Proposition 1 plus Lemma 1: the two models disagree only on next-states in the open interior, which lies outside $\mathcal{R}$. (ii) Imagined rollouts from an outside state under either model freeze at the same first-annulus landing — the models agree on A and outside, and by Lemma 1 (applied to imagined paths, which use the same integrator and step bound) imagination never produces an interior query where they differ. Every candidate action sequence therefore receives the same imagined return under both models; a deterministic planner selects the same action at every real step, and the real environment does the rest. □

Measured *bitwise*, as designed: paired-seed MPC episodes on f versus $\hat{f}_{\text{fill}}$ are identical in return, final state, and contact, seed for seed.

Remark 4 (the triad)*.* Proposition 2 is the cleanest statement of a three-way split this paper keeps separating: *certifiability*, *correctness*, and *consequence* are three different things. At $\gamma = 0$ the filled-disc artifact is wrong, certified, and costless; $\gamma > 0$ continuously converts the same wrongness into $(1-r(\gamma))^N$-gated danger — $E(f)$ shrinks and $\hat{f}_{\text{fill}}$ exits it. One knob walks one artifact through all three regimes (Section 4).

3.1 The γ-curves and the query lower bound

Fix one probability space for all γ: a single i.i.d. action sequence and initial state drive the dynamics at every channel width (common random numbers), so that $\gamma_1 < \gamma_2$ gives nested mode regions $A(\gamma_2) \subseteq A(\gamma_1)$ with difference slivers $D = A(\gamma_1) \setminus A(\gamma_2)$. Write $r(\gamma)$ for the per-rollout contact rate and $r_{\text{int}}(\gamma)$ for the interior-entry rate.

Lemma 2 (divergence localization)**.** *Under the coupling, the γ_1- and γ_2-trajectories coincide up to (and excluding) the first step whose landing falls in D; hence for any trajectory event, $|P_{\gamma_1} - P_{\gamma_2}| \leq P(\text{some landing in } D)$.*

Proof. Before that step every landing is either outside $A(\gamma_1)$ (both trajectories free, same next state) or in $A(\gamma_2)$ (both freeze at the same previous position); induct. The bound is the coupling inequality. □

Proposition 3 (r is nonincreasing in γ)**.** *Contact events are nested pathwise:* $\text{fire}(\gamma_2) \subseteq \text{fire}(\gamma_1)$*, so $r(\gamma_2) \leq r(\gamma_1)$.*

Proof. If the γ_2-trajectory first fires at step t with no D-landing before t, Lemma 2 makes the trajectories agree through t and the landing lies in $A(\gamma_2) \subseteq A(\gamma_1)$; if some D-landing occurs at $s \leq t$, a landing in D *is* a landing in $A(\gamma_1)$. (0 violations in 44,000 CRN checks.) □

Proposition 4 (continuity of the γ-curves, with explicit modulus)**.** *For every event determined by the trajectory — in particular for $q = r$ and $q = r_{\text{int}}$ — and all $0 \leq \gamma \leq \gamma' \leq 2\pi$ with $\varepsilon = \gamma' - \gamma$,*

$$|q(\gamma) - q(\gamma')| \;\leq\; h \cdot \sqrt{r_{\text{out}}\, \varepsilon / (\text{gain} \cdot dt^2)} \;\approx\; 1033\sqrt{\varepsilon} \quad \textit{at the defaults,}$$

so both curves are uniformly Hölder-$\frac{1}{2}$ on $[0, 2\pi]$; in particular $r_{\text{int}}(\gamma) \leq 1033\sqrt{\gamma} \to r_{\text{int}}(0) = 0$.

Proof. The one-step proposed landing from any state is *exactly* uniform on a circle of radius $R_L = \text{gain} \cdot dt^2$ centered at a state-determined drift point: the integrator gives $x' = x + (1 - \text{drag}\, dt)\, v_x\, dt + \text{gain} \cdot dt^2 \cos\phi$ (same for y) with $\phi = \pi a / a_{\max}$ uniform on $[-\pi, \pi]$ when a is uniform. Each of D's two slivers lies in a strip of width $r_{\text{out}}\varepsilon/2$ around its bisector line, and a circular-uniform law gives a width-w strip mass at most $\sqrt{w/(2R_L)}$ (the arcsine of a length-w/R_L interval; the worst case is tangency, where the bound is attained up to the factor $\pi/\sqrt{2}$). Condition on the state, apply this per step and sliver, and sum over h steps via Lemma 2. The exponent $\frac{1}{2}$ is not pessimism: a drift center at distance exactly R_L from a sliver line has single-step hitting probability $\geq \sqrt{w/R_L}/\pi$, and the measured divergence probability scales as $\varepsilon^{0.30}$ over $\varepsilon \in [0.0125, 0.2]$ — the channel-mouth (funnel) states occupy this tangency band and saturate per episode — so a linear bound on the coupling's divergence term is refuted at the measured scale. Full statements, the linear off-tangency refinement, and machine checks are in the repository's theory notes (Lemmas S/A/W, Theorems T4/T4′). □

Proposition 5 (direct interior entries are pathwise monotone)**.** *Call an entry* direct *if the trajectory never lands in $A(\gamma)$ before its first interior entry. For $\gamma_1 < \gamma_2$,* $\text{direct}(\gamma_1) \subseteq \text{direct}(\gamma_2)$*; the direct component of r_{int} is nondecreasing.*

Proof. A direct-at-γ_1 trajectory's pre-entry landings avoid $A(\gamma_1) \supseteq A(\gamma_2)$; by the Lemma 2 induction it is unchanged under γ_2 and still direct. □

Corollary 2 (monotonicity up to the funnel defect)**.** *Write $r_{\mathrm{int}} = d + f$, splitting entries into direct and funnel-assisted. For $\gamma_1 < \gamma_2$, $r_{\mathrm{int}}(\gamma_2) \geq r_{\mathrm{int}}(\gamma_1) - f(\gamma_1)$; in particular any violation of M1 (r_{int} nondecreasing) or M2 ($r_{\mathrm{int}}(\gamma) \leq r_{\mathrm{int}}(2\pi)$) is at most the funnel mass at the smaller gap, and is exactly zero where $f = 0$.*

Proof. $r_{\mathrm{int}}(\gamma_2) \geq d(\gamma_2) \geq d(\gamma_1) = r_{\mathrm{int}}(\gamma_1) - f(\gamma_1)$, the middle step by Proposition 5. At $\gamma = 2\pi$ there is no mode, so $f(2\pi) = 0$ and $r_{\mathrm{int}}(2\pi) = d(2\pi)$. □

Remark 5 (the defect is measured, and small). Full monotonicity admits *no* pathwise proof: probe seed 50543 enters the interior at $\gamma = 0.4$ but not at $\gamma = 0.6$ (nor at 2π). Corollary 2 explains rather than merely records it — Proposition 5 forbids a direct violation, so the counterexample must be funnel-assisted, which it is. It also bounds the damage. Measuring the defect over 50,000 common-random rollouts per gap: the funnel mass peaks at 22/50,000 around $\gamma \in [0.6, 0.9]$ and vanishes at both ends (nothing to enter at $\gamma = 0$; nothing to freeze against past saturation), giving a certified slack of 6.7×10^{-4} against an effect size of 1.12×10^{-2} — a factor of 17, with zero empirical violations and Proposition 5 confirmed across 550,000 pathwise comparisons. So M1 and M2 hold as theorems up to a defect an order of magnitude below the phenomenon; for $\gamma \geq 3.2$ the funnel mass measures 0/50,000 (a censored zero, Wilson upper 7.7×10^{-5}), so there the defect is at most that bound — exactness itself holds wherever $f = 0$ (Corollary 2(c)), which a censored zero bounds but does not prove. What remains open is an *a priori* bound on the funnel mass; the one pointwise estimate that would have given unconditional monotonicity was *refuted* by a freeze-rescue effect (91/91 CI-separated violations). Nothing below load-bears on M1/M2 (Corollary 3).

Proposition 6 (positivity: $r_{\mathrm{int}}(\gamma) > 0$ for every $\gamma > 0$, facing channel)**.** *At the frozen defaults with the channel facing the start, $r_{\mathrm{int}}(\gamma) > 0$ for every $\gamma > 0$.*

Proof (witness tube, compact). Condition on $|y_0| \leq \eta(\gamma) := (3.5/8)\gamma \wedge 0.4$ (an event of positive probability under the uniform start). The constant action $a \equiv 0$ drives the state due east along $y = y_0$; a landing inside the band at radius $d \geq r_{\mathrm{in}}$ has angular offset θ from π with $\sin\theta = |y_0|/d \leq |y_0|/r_{\mathrm{in}} \leq \gamma/8$. For $\gamma \leq \pi$, Jordan's inequality gives $\gamma/8 < \gamma/\pi \leq \sin(\gamma/2)$, and since both angles lie in $[0, \pi/2]$ this yields $\theta < \gamma/2$; for $\gamma > \pi$ the bound is immediate, $\theta \leq \pi/2 < \gamma/2$. The landings lie in the channel, so the witness path is freeze-free with clearance $c(\gamma) > 0$ from $A(\gamma)$ and first lands past the band at depth $< r_{\mathrm{in}} - c$. On the freeze-free tube the h-step flow map is Lipschitz in the action sequence (explicit constant L_h by the linear drag recursion), so every action sequence within $\rho = c/(2L_h)$ of 0 in sup norm still enters; that action tube has probability $\rho^h > 0$ (machine-checked witness: `test_positivity_witness_tube`). □

Corollary 3 (the knob is theorem-backed)**.** *$r_{\mathrm{int}}(0) = 0$ exactly (Lemma 1); r_{int} is continuous (Proposition 4); $r_{\mathrm{int}}(\gamma) > 0$ for every $\gamma > 0$ (Proposition 6). The γ-knob re-opens identifiability* continuously from an exact zero*; the instrument's claims need nothing from M1/M2.*

Positivity is also where sample size earns its keep: 30,000 rollouts measure 3 and 12 interior entries at $\gamma = 0.05$ and 0.1 (rates 1.0×10^{-4} and 4.0×10^{-4}; `results/ring2d_rarity_sweep.json`, unit rollout) — confirming Proposition 6 at gaps where the witness tube's own bound is astronomically small, and where any calibration of a few hundred rollouts would typically read 0, indistinguishable from the theorem's exact zero at $\gamma = 0$. Zero is a theorem only at $\gamma = 0$; everywhere else a printed zero is a sample statement.

Proposition 7 (query lower bound)**.** *Let the planner select, at each real step, the first action of an imagined argmax over a candidate set $\mathcal{C}$ under the blind model. Assume* (RG) *every candidate* *whose imagined path enters the phantom basin $B(c, r_0)$ out-scores every candidate that stays outside $B(c, r_{\text{out}})$, and* (C) $\mathcal{C}$ *contains an entering candidate. Then the selected imagined path crosses the annulus (Lemma 1 applied to imagined paths), so the planner queries the disagreement region with probability* 1*: the companion paper's play-cost upper bound is tight-side-active on this instrument.*

Proof. By (C) an entering candidate exists; by (RG) no non-entering candidate is the argmax; the entering path passes from outside r_{out} to inside $r_0 < r_{\text{in}}$ and Lemma 1 places one of its steps in A. (RG) is *checkable, not behavioral*: at the frozen defaults it holds with margin over the visited-state envelope and is verified numerically before every arm, in the pre-registration style of the discrete paper; (C) is logged. Measured: blind contact rate 1.0 at every γ. $\square$

The impossibility grades split cleanly: $r_{\text{int}}(0) = 0$ is *exact* (a theorem, measured 0.0000 at $n = 4000$), whereas the *hidden* channel at $\gamma > 0$ has $r_{\text{int}} > 0$ but of tube order ρ^h — positive yet below measurement at our budgets. These are two different *grades* of impossibility, and Sections 4 and 6 show the gate and the planner cannot tell them apart while behaving identically on both. The second grade now has an explicit **steering witness** (`scripts/ring2d_steering_witness.py`): a scripted waypoint policy that knows the hidden channel enters the interior in 100/100 episodes at hidden $\gamma = 1.2$ *within the instrument's own horizon*, where the gate policy's measured rate is 0/100 — and the same controller at $\gamma = 0$ enters 0/100, as the theorem requires. Certifiability is relative to the query policy ρ: $E(f)$ is defined by ρ's reach, and a policy that knows the channel shrinks the gauge region. (At hidden $\gamma = 0.6$ the narrower corridor defeats the *hand* controller — but a 10-parameter search over the same waypoint family finds a controller entering 100/100 within the same horizon, machine-checked. Reachability is jointly a property of geometry, policy, and budget, and the policy axis is cheap to search: what the gate policy cannot reach in 400 rollouts, ten tuned parameters reach every time.) Under two checkable hypotheses — (RG), a reward gap making some channel-entering candidate dominate every non-entering one, and (C), candidate coverage — the blind planner's imagined argmax path must cross the annulus, so the planner queries the disagreement region with probability 1: the companion paper's play-cost upper bound is tight-side-active on this instrument. (RG) is verified numerically over the visited-state envelope before every arm, in the pre-registration style of the discrete paper; (C) is logged. Measured: blind contact rate 1.0 at every γ in calibration.

4 Mechanism without LLMs: three regimes, one knob

Before any LLM enters, the mechanism grid measures what the knobs do to a *fixed* pair of wrong artifacts: the mode-blind model (the ring deleted — the companion papers' omission) and the filled-disc model of Proposition 2 (the wrong-topology invention). Each cell of $\gamma \times$ channel $\times$ start runs 400 uniform-random rollouts (the gate-side falsifiability rates) and 16 paired-seed MPC episodes per artifact (the play side).

Three readings organize the grid. **First, the three-regime walk is real.** The same filled-disc artifact is unfalsifiable and costless at $\gamma = 0$ (disagreement rate 0, $\text{pc}_{\text{fill}} = 0$ — the bitwise theorem observed), falsifiable at rate $\sim 10^{-3}$/transition and costly through a facing channel (pc_{fill} 0.343/0.220), and instantly falsified from inside (disagreement 0.77–0.97). One artifact, three certification regimes, two knobs.

Second, an invented mode is exploited below random from inside. $\text{pc}_{\text{fill}} = 1.769$ at $\gamma = 0$, inside start: the filled disc hallucinates freezes everywhere near the lode the agent already sits on, so all imagined returns tie and the planner drifts *off* the reward. This is the dual of the

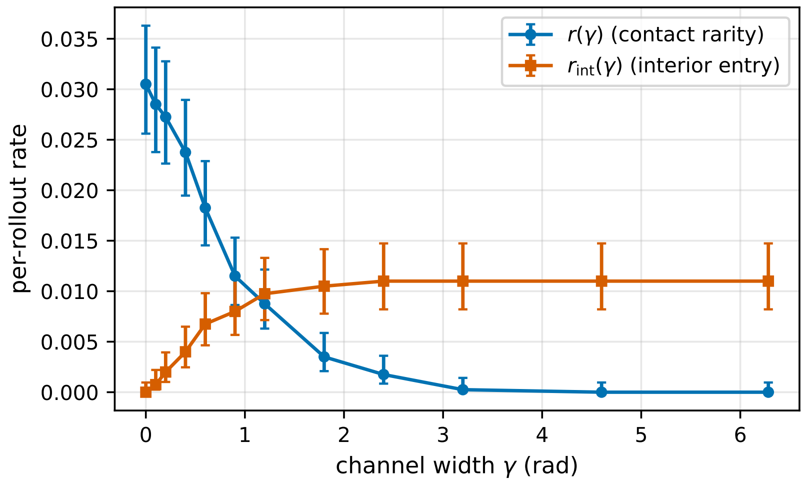

Figure 1: The γ-curves, measured with common random numbers (4,000 CRN rollouts per point; Wilson 95% CIs): the contact rarity $r(\gamma)$ is nonincreasing (pathwise theorem; 0 violations observed) and the interior-entry rate $r_{\text{int}}(\gamma)$ rises from its *exact* zero at $\gamma = 0$ (Lemma 1) — the two grades of impossibility are visible as the difference between a point pinned at zero by a theorem and a curve passing through measurably positive values.

companion papers' lure: an omitted mode pulls the planner into danger; an invented one repels it from value, just as destructively.

Third — the thesis, pre-LLM — policy-relative reachability beats topology. The hidden-channel rows are observationally *identical* to the closed ring at every tolerance (interior entries 0, disagreement 0, pc_{blind} 0.999, pc_{fill} 0.000) although the channel changes the free space's connectivity; with the *facing* channel — same topology — everything changes. What certificates and play see is the mode's position relative to the operative reach. Note the two grades of impossibility beneath the identical observations: the closed interior is reach-null *exactly* (a theorem), the hidden channel's interior is reachable at positive-but-unmeasurable rate; Section 6 returns to this pair with the danger curve.

5 LLM synthesis on the closed ring: three families

The synthesis arm runs the companion paper's protocol verbatim (Azure GPT-5.x mini and large, 20 seeds/cell, $N = 40$, $\varepsilon = 10^{-9}$, ≤ 5 refines) on four pre-registered cells of the closed ring: **A** (default prompt, outside start), **B** (region-first guidance, outside), **C** (guidance plus a per-seed topological summary of the sample's own contact evidence, outside), and **D** (the same summary, *inside* start). Because a filled disc at $\gamma = 0$ passes every metric (Proposition 2), code-level claims need more than the gate; the audit below is used for every artifact in the paper.

5.1 The behavioral audit (methodology)

Textual inspection of synthesized code misleads in both directions: a verbose "trap" comment can be measure-zero (behaviorally blind), and a plausible-looking region clause can be wildly misparameterized. The audit therefore classifies each artifact by what its `step()` *does*: probe it on an 81×81 state grid at two velocity slices, mark every deviation from the pinned integrator (the

γ	channel	start	r	r_{int}	disagree$_{\text{fill}}$	pc$_{\text{blind}}$	pc$_{\text{fill}}$
0.0	—	outside	0.0450	0.0000*	0.000000*	0.999	0.000*
0.6	facing	outside	0.0350	0.0075	0.000844	0.022	0.343
0.6	hidden	outside	0.0450	0.0000†	0.000000†	0.999	0.000†
1.2	facing	outside	0.0175	0.0125	0.000625	0.007	0.220
1.2	hidden	outside	0.0450	0.0000†	0.000000†	0.999	0.000†
0.0	—	inside	0.7325	1.0000	0.968500	0.000†	1.769
0.6	facing	inside	0.6875	1.0000	0.862969	0.000†	0.663
0.6	hidden	inside	0.6700	1.0000	0.854094	0.000†	1.769
1.2	facing	inside	0.6100	1.0000	0.774062	0.000†	0.351
1.2	hidden	inside	0.5975	1.0000	0.757500	0.000†	1.741

Table 1: The full mechanism grid. One wrong-topology artifact: unfalsifiable and harmless ($\gamma = 0$, the bitwise theorem), falsifiable and costly (facing channel), instantly falsified (inside start). The hidden channel is observationally identical to the closed ring although it changes β_1. Rates over 400 rollouts (r, r_{int}; the JSON carries per-cell Wilson CIs, e.g. $r = 0.045 \in [0.029, 0.070]$) and 32,000 transitions (disagree$_{\text{fill}}$); a zero cell has 95% Wilson upper bound 0.0095 (rollout rates) and 1.2×10^{-4} (transition rates) — every qualitative separation in the table exceeds its CI by orders of magnitude. pc over 16 paired MPC episodes. *exact — r_{int} and the interior disagreement by Lemma 1 (no sampled transition can land inside), pc$_{\text{fill}}$ bitwise by Proposition 2; †censored — no occurrence in that sample, content is the stated upper bound (episode-rate zeros: 0/16, Wilson upper 0.19).

artifact's effective freeze/mode set), and classify that set's shape by angular coverage around the ring center, interior fill, and boundedness: `blind` / `point` / `vdep` (velocity-conditioned) / `arc` / `loop` (hollow, bounded) / `disc` (filled) / `complement` (hollow, unbounded) / `fill-unbounded` (half-plane-like). The classifier is oracle-tested: artifacts constructed with a known class by construction (a hollow annulus, a filled disc, a half-plane, a velocity cap, a measure-zero trap, . . .) must be labeled correctly before the classifier touches real artifacts.

Two derived readings recur. **Blind-reference gates**: a terminal gate is only interpretable against the pure integrator's gate on the *same* evidence (≈0.999 outside, 0.971 inside) — the empirical gate bands are: correct ≈1.0; blind ≈0.97; posed-but-mis-parameterized hollow structures 0.5–0.9; interior-filling structures 0.02–0.4 (they freeze states the evidence shows moving). A posed region with wrong parameters scores far *below* blind — so "mean gate fell" can mean *more* mode-posing, not less, and reading terminal gates without the reference invites exactly the misreading we document (and correct) in Section 6. **Per-patch coverage**: the same mask measures whether an artifact encodes a *specific* region (e.g., the seen versus unseen patch of the companion paper's bi-modal instrument, where the audit verified the published hand inspection at 74/76 integrator exactness and 0 seen-patch coverage).

The identifiability event is family-independent. Every mode-absent sample — probability $(1-r)^{40} = 0.28$ at the defaults, with $r = 0.0312$ measured at 30,000 calibration rollouts (`results/ring2d_rarity_sweep.json`; realized: 6/20 seed blocks, the unit both sizes share) — yielded a certified, fully mode-blind artifact, exploited at play_cost ≈ 1.12 with contact rate 1.0, in every family tested: GPT-5.x (both sizes, full arm 20/20 clean in both), Qwen (HF router, spot check), and Claude (agent-relayed, spot check). The play number is bit-identical across families because the blind artifact is the same program: identifiability is a property of the sample, not the model — the companion papers' law, now on a mode with an inside.

cell	prompt / start	pass	present	blind ref	terminal classes (behavioral)
A_{lg}	default / out	6/20	14	0.999	fill-unbounded 13, blind 7
A_{mi}	default / out	6/20	14	0.999	blind 15, arc 2, disc 2, fill-unb. 1
B_{lg}	region / out	8/20	14	0.999	blind 16, arc 2, disc 1, point 1
C_{lg}	tda / out	9/20	14	0.999	blind 19, vdep 1
C_{mi}	tda / out	12/20	14	0.999	blind 17, arc 2, point 1
D_{lg}	tda / inside	1/20	20	0.971	arc 5, point 4, blind 4, disc 2, compl. 2, loop 2, vdep 1
D_{mi}	tda / inside	0/20	20	0.971	disc 5, loop 5, arc 5, blind 4, compl. 1

Table 2: GPT-5.x on the closed ring ($\gamma = 0$; lg = large, mi = mini; the two sizes share seeds, so their samples — and the mode-present counts — are identical by construction, not independent replications; `results/continuous_synthesis_ring2d_{mini,large}_gap0{,_pv-tda,-in_pv-tda}.json`, the seed block as the unit). "Blind ref" is the pure integrator's gate on the same evidence: the interpretive baseline for terminal gates. Every gate pass in A–C is a mode-absent blind artifact (exploited at play_cost $\approx$ 1.12) or a measure-zero point fit; the one D pass is the gauge-free complement form. Cross-family spot checks (3 seeds each): Qwen — both mode-absent seeds certified blind and exploited at the same 1.116, the mode-present seed refused by the gate; Claude (agent-relayed) — same A-cell outcome bit-for-bit, and 3/3 D-cell repairs via the same complement form as the GPT passer.

Held-out replication: independent-gate acceptance coincides with that gate's own mode-miss, 156/156**.** Re-scoring every committed artifact on an independent gate block (a disjoint 40-rollout stream per seed; `scripts/heldout_gate_audit.py` under its `ring2d` scope, output `results/heldout_gate_audit_ring2d.json`: 783 artifacts over 37 campaigns, the thin-neck synthesis cells of Section 6.1 included) turns the identifiability law into a held-out measurement. On the incomplete arm restricted to artifacts whose *training* sample missed the mode, held-out acceptance coincides with "the independent gate's own block also missed the mode" in 156/156 artifacts — an exact identity checked per artifact, over 91 disjoint seed blocks (the two sizes share their blocks by construction), zero off-diagonal — and the two-factor prediction $(1-r)^{N_{\text{train}}+N_{\text{gate}}}$ lies inside the Wilson 95% interval of the measured accepted-and-blind rate in every one of the 37 campaigns (each interval over that campaign's own disjoint, independent seed blocks — never pooled across sizes), the hidden-channel cells included — audited under the closed ring's r, which their own 30k calibration equals to five decimals. The same audit measures the converse: of 214 in-sample gate passes, 121 fail the independent gate, and every one of the 121 fails *at a mode contact* — an in-sample pass is training-set consistency, and what it omits is exactly the mode. Acceptance — by the training sample or by an independent block — is a sample event on this instrument in both directions, measured on held-out evidence.

None of the three tested families repairs the ring from outside evidence. Across A–C, zero mode-present artifacts pass the gate with the mode genuinely encoded. The behavioral audit is what makes this claim sharp: the gate-passing mode-present artifacts in B and C are *textual point fits* — integrator plus a comment hypothesizing a tiny localized trap and an exact-coordinate snap at 10^{-12} tolerance — whose freeze-mask is empty: behaviorally blind (wall_blindness = 1.0), exploited like the pure blinds. The held-out audit supplies the class's sharpest specimen: one trap (a hidden-channel cell, observationally the closed ring) freezes on exact `==` equality with its sample's single contact state, and the hardcoded coordinate sits 2 ulps from the same trajectory under a different `libm` — its stored in-sample 1.000 is a property of the last bit of sin / cos on the machine that ran the campaign, and the independent gate rejects it on every platform (`results/heldout_gate_audit_ring2d.json`, `train_reproduction_check`). This is Corollary 1 operating: from outside, ring and disc evidence are pathwise identical, so the honest summary can only report the reachable arc, and it does not pose the hole. Notably, richer prompting made the *terminal* attempts less geometric, not more: the default prompt's failing terminals are half-plane templates (13/14 mode-present, large), while region/summary guidance pushed terminals toward point fits and blinds (16–19/20).

From inside, repair is parameter-bound, not structure-bound. The D cell inverts the failure mode. With inside evidence and the loop-reporting summary, models *pose* geometric mode structure in most cells — mini: 5 filled discs, 5 hollow loops, 1 complement, 5 arcs of 20 — yet 0/20 pass (large: 1/20), and the terminal gates average 0.56, far *below* the 0.971 blind reference: a posed region with wrong parameters over-freezes states the evidence shows free. The bottleneck is not posing the topology (the guidance says loop and the models comply) but pinning its parameters to 10^{-9} from landing evidence. A dose curve makes this precise: re-running the D cell with $8\times$ the evidence ($N \in \{40, 80, 160, 320\}$ rollouts, 20 seeds each) yields 0/20 gate passes at *every* dose, with the posed structures' median inner-radius error pinned at 0.5 throughout — that is, the median artifact poses $r = 4.0$, a *round number*, whether it has seen 40 rollouts or 320. The parameter is not estimated from the evidence; it is guessed from the template library, so evidence dose cannot buy repair — the synthesis-side twin of Section 8's dose-independence.

The separator ablation: a square ring. Replacing the Euclidean norm by the Chebyshev norm turns the annulus into a square ring (zero curvature, corners) while the crossing lemma survives verbatim (Chebyshev distance is 1-Lipschitz with respect to Euclidean steps; reach-null and the bitwise Proposition 2 equivalence re-verify on the square). Every quantitative finding is norm-invariant: mode-absent samples yield certified blind artifacts exploited at 1.117; inside repair is 0/20 with structures posed in 15/20. The shape of what is posed is not: of the 12 posed closed structures on *square* evidence, 11 are round (freeze-boundary corner ratio ≈1.0 against the square's $\sqrt{2}$) and 12/12 are written with `hypot` — zero use the truth's `max`/`abs` form. The companion paper's bidirectional template prior, measured there on small patches, reaches the enclosing separator itself: the topology axis and the template-prior axis compose, and the gate-certified artifact's *shape* comes from the prior even when its *topology* comes from the guidance. The D failures are not template-prior collapse: the behavioral audit classifies them as right-structure-wrong-parameters, which is a different failure mode.

How the one passer passed. The single gate-certified D recovery (large, 1/20) freezes exactly when the post-integration landing has $\|p - c\| \geq 3.5$: the *disc-complement* form. It is certified perfect on its own gate sample (gate 1.0, blindness 0.0, play_cost 0.0) although its outer boundary is wrong — from inside, the probe freezes at the inner boundary and never reaches the outer one, so the complement is indistinguishable from the annulus on $\mathcal{R}$: the outer boundary is **gauge** (Proposition 1), and the artifact's comment even cites the summary's "closed loop ($\beta_1 = 1$)." Its single load-bearing parameter is the true inner radius — a *round number* anchored at the center that the contract's reward section already names. The strongest cross-family repairer (Claude, agent-relayed context-free, 3/3 D repairs) wrote exactly the same form. Gate-pass from inside therefore measures a conjunction — *(reachable-equivalent structure)* × *(exact-parameter guessability)* — not topology-understanding alone; Section 6 exploits this by moving the true parameters off round values.

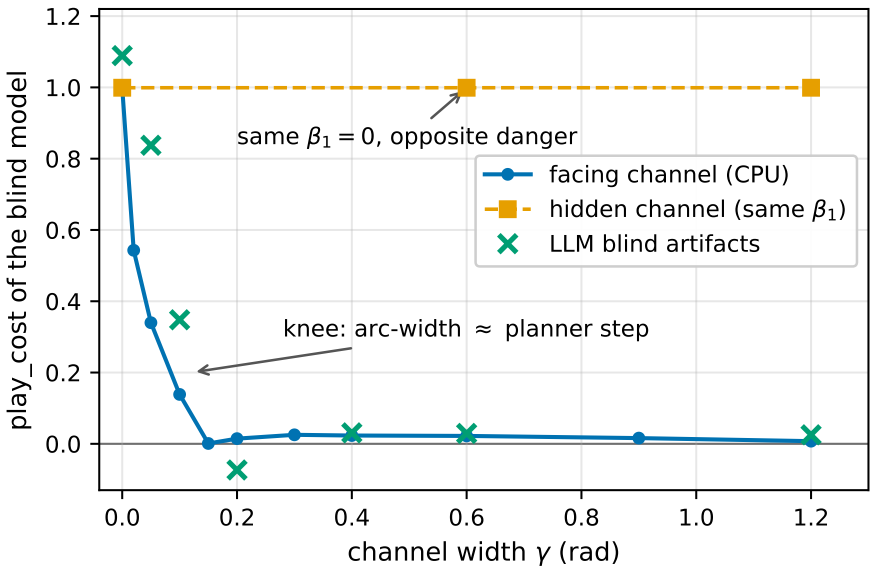

Figure 2: **H1: danger is topology relative to reach.** The blind model's exploitation collapses over a knee at $\gamma \approx 0.1$ when the channel faces the start (blue: dense CPU curve, 16 paired MPC episodes per point; green: the LLM arms' exploited blind artifacts, identical in all three model families) — while the *hidden* channel (orange) holds full danger at the same γ and the same $\beta_1 = 0$. The knee sits where the channel's arc width admits the planner's step.

6 The open-ring arm: danger is topology relative to reach

The registered arm sweeps γ with three pre-registered hypotheses: **H1**, the blind model's exploitation collapses with a facing channel and persists with a hidden one (danger is reachability, not β_1); **H2**, posed artifact structure tracks the guidance's reported topology (Section 8); **H3**, gate-pass from inside stays ≈ 0 at every γ because the channel-edge angles $\pi \pm \gamma/2$ are not round-guessable. Cells: outside/default at $\gamma \in \{0.05, 0.1, 0.2, 0.4, 0.6, 1.2\}$ facing (20 seeds mini) and $\{0.6, 1.2\}$ hidden (10); inside/summary at $\{0.2, 0.6, 1.2, 1.8, 2.4\}$ (20–30); large robustness cells at the critical points; a dense CPU danger curve (16 paired MPC episodes per γ); and Claude relay spots at the knee and at wide γ.

γ	0.0	0.02	0.05	0.1	0.15	0.2	0.4	0.6	1.2
$\mathrm{pc}_{\mathrm{blind}}$ (facing)	0.999	0.543	0.340	0.139	0.001	0.015	0.023	0.022	0.007
contact rate	1.00	0.94	0.56	0.44	0.12	0.06	$0.00^{\dagger}$	$0.00^{\dagger}$	$0.00^{\dagger}$

Table 3: H1, the danger curve (CPU, 16 paired MPC episodes/point): the blind model's exploitation collapses over a knee at $\gamma \approx 0.1$ — channel arc-width $\gamma\, r_{\mathrm{in}} \approx 0.35$, comparable to the planner's step. The *hidden* channel holds pc $= 1.116$ at $\gamma = 0.6$ *and* 1.2: same β_1, opposite danger. LLM-arm confirmations at $\gamma \in \{0.1, 0.6\}$ reproduce the curve in both model sizes and in the Claude relay (0.348 at the knee, all three families); Figure 2 plots all three series. †censored: 0/16 episodes, Wilson upper 0.19.

H1 confirmed — with the clincher (`results/continuous_ring2d_open_sweep_summary.json`, one seed-block cell per (γ, arm, size)). The facing curve collapses over a knee at $\gamma \approx 0.1$–0.15; the LLM cells reproduce it on their exploited blind artifacts in both sizes and in the Claude relay (0.348 at the knee in all three families — the play number is the blind program's, family-independent by construction). The *hidden* channel at the same γ holds play_cost = 1.116 — full strength, unchanged from the closed ring. Same $\beta_1 = 0$, opposite danger: what collapses the danger is the competent planner's reachability of the omission, not the topology change. The mechanism is the gate quotient on the play side: as the channel opens where the planner drives, the phantom stops being phantom — the blind plan (straight at the lode) becomes *executable in the truth*, so blind and truth agree along the operative path exactly as $E(f)$-members agree on $\mathcal{R}$. The knee sits where the channel's arc width ($\gamma\, r_{\text{in}} \approx 0.35$ at $\gamma = 0.1$) admits the planner's step.

H3 confirmed — certified passes are not repairs. Inside gate-pass rates across γ: 1/40, 0/20, 0/40, 0/20, 1/30, 1/40 — essentially zero everywhere, both sizes, with terminal gates far below the blind reference (structures posed, parameters unpinnable: the channel edges are not round numbers). The sharpest datum is the strongest repairer's: at $\gamma = 2.4$ the Claude relay *reconstructs* "ring $r = 3.5$, gap 2.4 rad" from the sample and passes the gate in 2/3 seeds — with mode blindness 1.0 and play_cost 0.0. The passing models match the sample through wrong parameters, fail the mode probes, and are harmless only because the wide-open ring no longer obstructs anything. Certification, correctness, and consequence come apart again (Remark 4), now with real synthesis. Held-out re-scoring adds what the inside failures are made of: 237/320 inside-start artifacts fail the independent evaluation *off* the mode as well (accuracies down to 0.05), against 87/343 for outside starts (`results/heldout_gate_audit_ring2d.json`, unit artifact) — the loop-evidence regime produces globally wrong models, not clean integrators missing a clause.

The identifiability axis moves independently alongside. As γ grows the ring shrinks and the mode-absent rate rises ($6 \to 16$ of 20 over the sweep): the synthesis axis (does the sample contain the mode?) and the play axis (does the omission obstruct the planner?) are separate knobs even within one instrument — the theme Section 9 pushes to its extreme.

6.1 The metric converse: a thin neck, topology held fixed

The γ-channel reopens the interior *topologically*. Its converse thins the band from outside inside a 0.3-rad sector (r_{out} dips to r_{in} + neck; the hole is invariant): $\beta_1 = 1$ and the band still separates the plane in the continuum at every neck > 0, but interior entry now requires a *single step longer than the neck* — the local crossing lemma, machine-checked in Lean beside Lemma 1: a mode set containing the thin annulus $[r_{\text{in}}, r_{\text{in}}+w]$ with $w > \Delta$ seals the hole whatever its shape. With $\Delta = 1.0$ the knob crosses Lemma 1's metric hypothesis while the topology never moves, and reachability below the threshold is *exhibited*, not sampled: a deterministic constant-thrust witness leaps the 0.5 neck (one 0.547-step from $d = 4.03$), and the same 40-witness family is blocked at neck ≥ 1.0.

neck (facing)	r	r_{int}	dis_{fill}	pc_{blind}	pc_{fill}
closed	0.0312	0^*	0^*	0.999	0.000^*
0.1	0.0227	0.0020	4.3×10^{-4}	0.451	0.594
0.2	0.0248	1.7×10^{-4}	$0^\dagger$	0.962	0.573
0.4	0.0255	$0^\dagger$	$0^\dagger$	0.971	0.500
0.6	0.0262	$0^\dagger$	$0^\dagger$	0.962	$0.000^\dagger$
0.8	0.0269	$0^\dagger$	$0^\dagger$	0.976	$0.000^\dagger$
1.2	0.0292	0^*	0^*	0.993	0.000^*

(30,000 rollouts, 320,000 disagreement transitions and 16 paired MPC episodes per cell, `results/ring2d_thin_neck.json`; design and readings pre-registered in the repository before the run. *exact — the closed row by Lemma 1 and Proposition 2, the neck-1.2 row by the local lemma at $\Delta = 1.0 < 1.2$, imagined steps included; †censored — 0 occurrences in that sample, Wilson 95% uppers 1.3×10^{-4} (rollouts), 1.2×10^{-5} (transitions), 0.19 (episodes).)

Three findings. **Hidden necks are bit-identical to the closed ring** at all six thicknesses (the same r to the last float, 0 entries, $\mathrm{pc}_{\mathrm{blind}} = 0.999$): reach, not geometry, a third time on this instrument. **The planner leaks only at the thinnest neck**: $\mathrm{pc}_{\mathrm{blind}}$ collapses to 0.451 at neck 0.1 (blind contact rate 0.56 — the blind planner escapes through the neck in half the paired episodes) and recovers to 0.96–0.99 from neck 0.2 up, because one landing in the thin band pins the episode (freeze zeroes the velocity, and from rest the next step is $R_L = 0.03$): threading needs the landing phase to skip a band the realized approach rarely skips. **Certified-and-costly, on the metric axis**: the filled-disc model can disagree with truth on a sampled transition only via a leap — 139/320,000 transitions at neck 0.1, 0/320,000 at every thicker neck — yet $\mathrm{pc}_{\mathrm{fill}} = 0.59/0.57/0.50$ at neck 0.1/0.2/0.4, with fill contact rate 0.00: the cost is plan divergence, because the *planner's imagined candidates* leap (imagination reaches speed > 4) where the random gate's rollouts never do (arrival speeds ≈ 1.5). At neck $\in \{0.2, 0.4\}$ the wrong topology is behaviorally indistinguishable from truth across 320,000 sampled transitions and 30,000 rollouts while costing half the play margin — the companion paper's certified-region/query-mass gap, reproduced with the topology knob held fixed. The gate certifies where it looks; the planner leaps where it plans.

The loop does not write the neck: 0/120 artifacts pose any angular structure. The synthesis analogue of posing the hole is posing the dip, and at neck $\in \{0.1, 0.2, 0.4\}$ (facing, incomplete arm, both sizes, 20 seeds per cell; `results/continuous_synthesis_ring2d_{mini,large}_gap0-nk{0.1,0.2,0.4}.json`, scanned by `results/ring2d_neck_synthesis_scan.json`) no artifact does either: 0/120 contain an angular term of any kind and 0/120 pose even a *uniform* band — the conditionals that appear are one-sided discs at the reward radius, velocity thresholds, and three exact-coordinate point fits of the family above, one of which (freezing on `==` equality with its training sample's single contact state) reached in-sample 1.000 on the platform that synthesized it and fails the independent gate `mode_only`. Evidence is not the bottleneck: the two seed blocks at neck 0.1 whose training samples contain leap-through transitions (16 and 12 interior landings; the sizes share their blocks by construction) are answered by a reward-threshold freeze and by free flight — shown the metric hole, the loop writes no geometry at all. All 45 gate passes across the six cells are wall-blind, and the held-out audit absorbs the cells without strain (the 37/37 and 156/156 above). The regime above — certified relative to the gate's sampled transitions, costly in play — therefore carries a synthesis-side converse: at necks the gate cannot distinguish from the closed ring, the accepted artifacts are the same blind integrators as everywhere else, admitted by the same two-factor law.

7 Mitigation against an enclosed mode: a covering law

The companion continuous paper closed its loop with a planner-side defense: *distrust-region replanning*. After each real step the planner compares the model's prediction to the observation; a mismatch records a point *fence* at the refuted prediction, and imagined rollouts truncate when they cross near a fence — on the 1D clamps and the small convex patches this collapses the blind planner's exploitation to a bounded first-contact transient, at zero cost when the model is correct. The same module, verbatim (`pos_dims` $= (0, 1)$, the patch-calibrated fence radius $\varepsilon_f = 0.5$), meets the ring:

cell	ε_f	pc_{model}	$\text{pc}_{\text{mitigated}}$	contact	fences/episode
closed ring, outside, blind	0.5	0.999	1.003	1.00	3.8
closed ring, outside, blind	1.0	0.999	1.005	1.00	2.3
closed ring, outside, blind	2.0	0.999	0.742	1.00	1.0
hidden $\gamma = 0.6$, outside, blind	0.5	0.999	1.003	1.00	3.8
closed ring, *inside, filled*	0.5	1.769	**1.769**	1.00	18.6
nerve fence, episodic	0.5	0.999	0.957	1.00	2.0
nerve fence, persistent	0.5	0.999	**0.058**	0.06	0.1
freedom patch (inside, filled)	0.5	1.769	**0.029**	—	80

Table 4: Distrust-region replanning on the ring (16 paired MPC episodes per cell; `results/continuous_mitigation_ring{,_eps1,_eps2,_nerve}.json` and `results/continuous_ring2d_optimism.json`, the paired episode as the unit). Point fences — the defense that collapsed the patch exploitation — do *nothing* at their calibrated radius, partially relieve only at a radius of the geometry's own scale, and are exactly inert against the invented mode while firing constantly (18.6 fences per episode, pc unchanged to three decimals). The hidden-channel row is identical to the closed ring, as Section 4 predicts. The last two rows are the constructive turn: a *nerve* fence (violations linked into tangentially-extended segments) still fails episodically, and collapses the exploitation once fences *persist* across episodes — two lessons total (episode 1), truth-equal returns ever after.

The failure is a covering law. A point fence is a zero-dimensional object; the ring's reachable boundary is one-dimensional (the west outer arc, $\sim$16 world-units at the defaults). Sealing a 1-dimensional boundary with ε_f-balls needs covering-number-many violations — boundary measure over fence radius — and the argmax planner concedes only 2–4 contacts per episode while re-routing through unfenced arc. The companion paper's "collapse decays with mode distance" was the small-boundary shadow of this law: on a patch of circumference $\sim$6 at $\varepsilon_f = 0.5$ a handful of fences suffices; on the ring they never do. Coarsening the fence to the geometry's own scale ($\varepsilon_f = 2.0$) buys partial relief here but is not a uniform remedy — that radius would swallow the patch instruments whole. The structural fix is dimensional — and it works, but only jointly with persistence. A *nerve* fence links violations into segments and extends them 3 units along the estimated boundary tangent (two violations 0.5 apart already fix the local direction). Episodically it still fails (pc 0.957): a single episode concedes only $\sim$2 lessons, and instrumentation shows the planner simply re-crosses the wall *in imagination* beyond the sealed corridor — the blind model lets any unfenced crossing through, so the agent slides along the boundary at one freeze per lesson. Persisting the fences across episodes (deployment monitoring rather than per-episode defense) changes the accounting: the boundary's covering cost is paid *once*. Measured: 2 violations in episode 1 (return 1.01), then **zero** violations and returns equal to the truth planner's in every one of episodes 2–16 (17.04 vs 17.04, ...); aggregate pc 0.999 $\to$ 0.058, contact rate 1/16. The mechanism is Proposition 1 run in reverse: the extended fence reconstructs enough of the boundary that the blind model *plus its fence memory* becomes planner-equivalent to the truth on the operative side — certification could not distinguish members of $E(f)$, and the defense engineers the planner back into it. This is theorem-backed under checkable hypotheses:

Proposition 8 (fence sufficiency)**.** *Suppose* (COV) *every segment of length* $\leq \Delta$ *crossing the reachable outer boundary arc of* A *passes within* ε_f *of the fence set, and* (RG-west) *the best non-*

crossing candidate's imagined return exceeds every crossing candidate's pre-crossing return plus its tie-break advantage (at the defaults: $\approx$*2 versus* $\leq$0.1*, checkable over the visited envelope). Then every crossing candidate truncates at or before its crossing; every non-crossing candidate scores identically under truth imagination and fenced-blind imagination (the models agree off the band, and truth imagination freezes only on crossings); and the argmax coincides. The mitigated planner* is *the truth planner:* play_cost $= 0$ *exactly.*

Episode 1 is the learning transient the proposition does not (and should not) cover: (COV) is established *by* its two lessons. And (COV) is policy-payable by three routes, measured head-to-head (all persistent): fence *geometry* (the tangential extension: 2 lessons, pc 0.058), *exploration* (a boundary-tracing probe collects 50 lessons in one episode — plus 11 patch-ups, since its 1.05-unit spacing left sub-ε_f gaps: the covering law bites the prober too — pc 0.123), and *passivity* (the planner's own contacts, 9 lessons spread over 16 episodes, pc 0.152 with occasional regressions). Persistence is the ingredient common to everything that works; geometry beats exploration beats waiting. The fence is itself an estimated boundary, i.e. the same object as Section 8's evidence summaries, and inherits the same resolution limits.

Against an invented mode, distrust is inert by construction. The filled-disc model from inside (the below-random cell of Section 4) hallucinates freezes everywhere, so every imagined future collapses to the current position and all candidate actions tie *before* any fence can matter; truncating trust in imagined value cannot help when the lie has already flattened imagined value. The fences fire constantly (18.6 per episode — the defense *detects* the lie every few steps) and change nothing (pc identical to three decimals). Distrust-based mitigation is one-sided in exactly the sense the danger is two-sided: it can stop the planner from *believing* false free space, but recovering value that the model hides behind false obstructions needs the *dual* certificate. We built it: **freedom patching** records, at each real step where the model predicted a freeze and the truth moved, a freedom point; during imagination, a model step that freezes within ε_f of a freedom point is replaced by the contract's own pinned integrator (mode-free, legitimately known to the pipeline). On the same cell this collapses the below-random exploitation at once: pc 1.769 $\to$ 0.029 episodic (0.021 persistent), with near-truth returns from *episode 1.* The asymmetry of the two defenses' costs is itself the finding: an invented mode *lies at every step* (80 freedom certificates per episode — it refutes itself constantly, so within-episode patching suffices), while an omitted mode lies *rarely* (2 lessons per episode — so its fence must persist to ever pay the covering cost). The dual result is theorem-backed the same way:

Proposition 9 (patch sufficiency)**.** *Let* $\hat{f}$ *freeze on a region* B *where* f *moves freely, and suppose* (CERT)*: every point of* B *visited by any candidate's imagined path from the current real state lies within* ε_f *of the freedom-certificate set. Then patched imagination equals truth imagination for every candidate (off* B *the models agree; on the part of* B *the freedom certificates cover, the patch substitutes the pinned integrator, which* is f *there), so the argmax coincides with the truth planner's and* play_cost $= 0$ *exactly.*

From inside, every step yields a certificate (the model lies each step — the lie rate is the occupation of the disagreement region under the operative policy), so (CERT) holds within one episode's prefix; the measured 0.029 is that prefix. The omitted-mode lure and the invented-mode repulsion (Remark 4's two wrongnesses) need *opposite* planner-side defenses; a defense calibrated for one is inert against the other; and each defense's price is set by its failure's *lie rate.*

8 The evidence sensor has finite resolution

The C and D cells' guidance includes a pre-registered *topological summary* of the sample's own contact evidence: an honest, shape-agnostic text (never naming a shape family, wording frozen before any run) reporting cluster count, bounding box, and a persistent $\hat{\beta}_1$ from a Rips complex on the deduplicated landing cloud (grid 0.05, cap 90 points, bars above 3× the median nearest-neighbor spacing). The summary is a *sensor*, and sensors have resolution: measured on regenerated inside evidence, the detector reports $\hat{\beta}_1 = 1$ for every $\gamma \leq 1.2$ — it bridges any channel narrower than about two arc-units at this density — flips seed-dependently at $\gamma = 1.8$, and is cleanly 0 at 2.4; the true β_1 is 0 for every $\gamma > 0$. The sweep therefore manufactures, at no extra cost, the experiment H2 needs: gaps where guidance and truth disagree, one gap where guidance disagrees with itself across seeds, and gaps where they agree.

The resolution limit is geometric, not budgetary. A factorial over the detector's budget and the evidence dose (`scripts/ring2d_sensor_resolution.py`: cap $\in \{30, 90, 270\}$ points × evidence $N \in \{40, 160\}$ rollouts × γ, five seeds each) shows the flip does not move: $\hat{\beta}_1 = 1$ at every $\gamma \leq 1.2$ under *every* budget and dose, 0 at 2.4 under all of them, with only the boundary gap 1.8 wobbling. The mechanism is Rips geometry, and it can be written as a law: to the resolution of the proved two-sided sandwich below, the detector reports $\hat{\beta}_1 = 1$ when $\sqrt{3}\rho - 2\rho \sin(\Delta\theta_{\max}/2) > \tau$ — the mean-radius display is the sandwich's collapse at $r_{\min} \approx r_{\max} \approx \rho$, and as a pointwise predictor it is measured, not exact — where ρ is the cloud's mean radius, $\Delta\theta_{\max}$ the *largest angular gap of the sample* (the channel or a subsampling gap, whichever is larger), and τ the detector's persistence threshold — the bar is born exactly at the largest gap's chord (proved: a winding cycle must contain an edge spanning every angular gap), cannot die before $\sqrt{3}\, r_{\min}$ (proved self-containedly: below that scale every triangle's boundary has winding zero, so winding obstructs filling), and dies by $2r_{\max} \sin(\theta^*/2)$ with an explicit θ^* (the filling lemma, proved: an explicit gap-aware spanning triangle of winding one plus a fan retraction exhibit the 2-chain that bounds the born *class* — no general filling machinery needed — and a pairing lemma transfers the sandwich from the class to the *bar* whenever exactly one bar spans $[2r_{\max} \sin(\Delta\theta_{\max}/2), \sqrt{3}\, r_{\min})$, a rank condition read off the barcode and true in every factorial row; $\theta^* \to 2\pi/3$ for dense samples, recovering the Vietoris–Rips circle constant (Adamaszek and Adams, 2017), and the measured death/$\bar{\rho} \in [1.70, 1.82]$ around $\sqrt{3} = 1.732$ is the radial thickness). The law is therefore *two-sided*: $\hat{\beta}_1 \geq 1$ is guaranteed when $\sqrt{3}\, r_{\min} - 2r_{\max} \sin(\Delta\theta_{\max}/2) > \tau$, and no persistent winding bar exists when $2r_{\max} \sin(\theta^*/2) - 2r_{\min} \sin(\Delta\theta_{\max}/2) < \tau$ (only a spurious non-winding class could still fire — never observed). On the factorial: the sandwich holds for 57/57 winding bars, the guaranteed bands have 0 violations (34 and 5 rows), and the pointwise law reproduces 78/80 rows, the 2 misses falling in the proved undecided band ($\gamma = 1.8$ boundary, subsampling-gap lottery). Worse, more evidence is counterproductive at the boundary: at $\gamma = 1.8$ the false-loop rate *rises* from 1/5 to 3/5 seeds when the dose quadruples ($n = 5$ paired seeds — directional, underpowered; `results/ring2d_sensor_resolution.json`, cap 90). The per-seed mechanism is that the denser subsample fills the channel-adjacent shells: on the two flipping seeds the spurious bar's persistence grows with the dose ($0.05 \to 0.50$ and $0.11 \to 0.43$), outrunning the threshold, which itself grows modestly (4/5 seeds) — the sensor gets *more confident* in the wrong topology as the data grows. Resolving a narrow channel requires a different filtration, not a bigger sample. A prototype confirms the direction and locates the real problem: censoring Rips edges that free trajectory segments properly cross restores specificity at every $\gamma > 0$ (19/20 versus 0/15 for plain Rips at $\gamma \leq 1.2$) at a sensitivity cost at $\gamma = 0$ (2/5 true loops lost). But edge deletion has a topological pathology with a clean characterization: a censored filtration stabilizes at the flag complex of the censorcomplement graph, so *never-fillable cycles* (infinite H_1 bars, impossible for plain Rips) are exactly that complex's H_1 — per-sample decidable by one finite computation. Running that certificate on all cells finds the pathology is real and *semantically ambivalent*: the same mechanism carries the true $\gamma = 0$ loops (their fills cross the certified-free hole, so the correct loop gets infinite confidence) and the prototype's single false positive at $\gamma = 0.6$, which turns out to be a structural never-fillable cycle, not a near-threshold bar.

Edge deletion is therefore the wrong primitive, and the principled replacement is *relative* homology of the pair $(\mathrm{VR}(X \cup Y), \mathrm{VR}(Y))$, contact evidence X against certified-free evidence Y. Two things make it the right object. It cannot produce infinite bars at all — both complexes are full simplices past the diameter, so every relative class dies — and by the long exact sequence its rank is $\operatorname{rank}\ker(H_0(\mathrm{VR}(Y)) \to H_0(\mathrm{VR}(X \cup Y)))$ whenever the union has trivial H_1: *the number of certified-free components that the contact evidence glues together*, computable by two union-finds rather than a matrix reduction. One instantiation detail is load-bearing: Y must enter as trajectory *polylines*, not as a point cloud. A free trajectory certifies passage between its own consecutive samples; discard that and the estimator reports separation almost everywhere, because a curve-dense contact cloud glues a scatter-sparse free cloud by density mismatch alone. With paths, the estimator is correct on 20/20 open-ring cells where the pre-registered detector scores 9/20, with no censoring and no infinite bars. At $\gamma = 0$ it does not fire, for two reasons we keep separate: with one-sided (inside) evidence the interior is reach-null, nothing outside is sampled, and separation is *unidentifiable by any estimator* — Proposition 1 asserting itself, with plain $\hat{\beta}_1$ scoring well there only because it answers a different question (the cloud's shape, not the separation it induces). With two-sided evidence the obstruction is different and, we think, more interesting: freeze semantics resets the velocity on contact, so after the first freeze the next proposed landing moves only $\text{gain} \cdot dt^2 = 0.03$ from rest and the trajectory *creeps along the face* instead of penetrating. Contact evidence therefore occupies two thin shells at the band's faces and never its interior, which caps the informative bar at the shell thickness while the persistence threshold grows with sample size as the shells fill in. Measured at 40/120/320 rollouts per arm: shells saturate at $[3.50, 3.71]$ and $[4.83, 5.00]$, bar length 0.160/0.197/0.253 against $\tau = 0.183/0.324/0.439$ — sub-threshold at every size, with the gap *widening*. So the registered $3\times$ calibration is not rescued by dose — the bar-to-threshold gap widens as the data grows; whether some other single calibration could recover $\gamma = 0$ while keeping the open-ring specificity is unmeasured. The same one-step landing law that makes the continuity modulus explicit (Proposition 4) is what caps the sensor here. Only the interleaving-stability theorem, which the pair formulation makes routine, remains. Nested boundaries break the detector a third way: with two concentric rings (the multi-chamber instrument) the middle-chamber evidence cloud is two concentric circles ($\beta_1 = 2$), and the pre-registered detector returns a seed-dependent *lottery* — $\hat{\beta}_1 = 0$ on 9/20 samples, 1 on 10/20, the true 2 on 1/20 — as Rips bridges between the circles merge or kill loops at this density. The synthesis side is correspondingly worse than the single ring: 0/20 gate passes and *zero* artifacts pose a nested (two-band) structure — 13/20 pose single arcs hugging one boundary. Certification peels one boundary layer per start placement (Proposition 1); the repair loop, at this depth, does not even *pose* layer two.

H2: the artifacts follow the sensor. Closed structures dominate while the summary says "closed loop" and all but vanish where it honestly says "arc" (Table 5: 1 closed versus 26 arc at $\gamma = 2.4$, both sizes); the crossover tracks the detector flip, not the true β_1. The within-gap contrast at $\gamma = 1.8$ — seeds of the *same* environment whose own summaries straddled the flip — is directional only (closed structures 3/27 in the $\hat{\beta}_1 = 1$ subset versus 2/33 in the $\hat{\beta}_1 = 0$ subset, $n = 60$: consistent in sign, not significant). What these measurements establish is: **posed topology tracks**

γ	0.0	0.2	0.6	1.2	1.8	2.4
closed structures (disc/loop/complement)	17	9	22	7	5	1
arc structures	10	5	6	9	34	26
n (cells; both sizes where run)	40	20	40	20	60	40
guidance $\hat{\beta}_1$	1	1	1	1	mixed	0

Table 5: H2: posed artifact topology tracks the *guidance's* $\hat{\beta}_1$, not the true β_1 (which is 0 for every $\gamma > 0$): closed structures dominate while the summary says "closed loop" and all but vanish where it honestly says "arc" — at $\gamma = 2.4$, 1 closed versus 26 arc across both sizes. n counts inside-start cells (mini at every γ; large twins at 0.0, 0.6, 2.4). The within-gap contrast at $\gamma = 1.8$, where seeds' own summaries straddle the detector flip, remains directional-only after doubling the cell to $n = 60$: closed structures 3/27 in the $\hat{\beta}_1 = 1$ subset versus 2/33 in the $\hat{\beta}_1 = 0$ subset — consistent in sign, not significant; the load-bearing H2 evidence is the cross-gap crossover.

the sensor's report, not the truth — the cross-gap crossover above. That an honest finite-resolution summary thereby *causes* wrong-topology certified models is the reading the data are consistent with but do not isolate: the cross-gap comparison varies the environment together with the summary, and the isolating intervention — randomizing correct against deliberately flipped summaries on the same committed evidence blocks, scoring posed topology and gate passage per seed block — is identified but not run. The repair loop's topology channel is, at minimum, no better than the sensor's resolving power at the operative sampling density.

The same law governs the sensor directly in n dimensions. On ShellField-n (Section 9), persistent homology of the *outside* contact cloud recovers β_{n-1} at *no* $n \in \{2..6\}$ — at low n the cloud is plentiful but traces only the reachable arc (signal clarity), at high n it starves as $r(n)$ collapses (0.18× down to 0.002× the Niyogi–Smale–Weinberger covering density (Niyogi et al., 2008), computed on the points the detector actually receives) — while the *inside* cloud recovers it cleanly at $n = 2..5$. The density bookkeeping matters: inside clouds *offer* 10–2700× the NSW floor in raw trajectory contacts, but the detector consumes a deduplicated subsample capped at 300 points — 24× the floor at $n = 2$ falling to 0.22× at $n = 5$ — so recovery held at and even below the naive floor (the NSW bound assumes i.i.d. uniform samples; a well-spread deduplicated subsample beats it), and at $n = 6$ only a 2-plane projection diagnostic was computed, which detects projected circles and is *not* evidence about β_5. Recovery is start-governed: reachability again, now for the sensor itself. We flag the program-level statement this suggests, as a conjecture and not a result: *the topology a certified repair loop can recover is bounded by the persistent-homology resolution of its contact evidence at the operative density* — a bound that NSW-style estimates make computable in advance, provided the operative density is the consumed subsample's, not the raw stream's.

9 Dimension as the rarity knob

`ShellField`-n keeps the two-lode geometry and places the freeze band on a spherical shell S^{n-1} around the phantom lode (both lodes in the first two coordinates, so n is the only knob); the action becomes a thrust vector $\vec{a} \in [-1, 1]^n$, norm-capped. Two facts scale in opposite directions. The gate-side rarity $r(n)$ collapses geometrically — so by $n \geq 3$ the identifiability event is near-certain: n **is a rarity knob**, and the mode is almost never in the sample. The play side does not collapse: truth-MPC still reaches the real lode at every $n \leq 6$ (the planner is not the bottleneck), and the blind model is exploited at full strength at every n (Table 7).

	outside start			inside start		
n	contacts	used (×NSW)	β_{n-1}?	contacts	used (×NSW)	β_{n-1}?
2	388	219 (17.4)	✗	33,702	300 (23.8)	✓
3	167	160 (2.5)	✗	45,697	300 (4.7)	✓
4	59	55 (0.18)	✗	54,170	300 (1.0)	✓
5	11	11 (0.008)	✗	59,961	300 (0.22)	✓
6	18	15 (0.002)	✗	63,949	300 (0.05)	proj. only

Table 6: Persistent homology of the ShellField-n contact cloud (20,000-rollout budget; alpha complexes; `results/continuous_shellfield_tda{,_inside}.json`, one cell per (n, start)). "contacts" counts raw trajectory contacts (dependent samples); "used" is the deduplicated subsample the detector receives (cap 300), and "×NSW" is *used* relative to the Niyogi–Smale–Weinberger i.i.d. covering bound. Outside evidence never recovers the shell's β_{n-1} — at low n the cloud traces only the reachable arc; at high n it starves — while inside evidence recovers it up to $n = 5$, at and below the naive floor. The $n = 6$ inside entry is a 2-plane projection diagnostic (projected H_1), not a β_5 computation. Recovery is start-governed.

What sets the collapse rate: solid angle. The mechanism is not variance concentration but direction. Reaching the shell from x_0 forces the displacement into the *tangent cone* to $B(c, r_{\text{out}})$: if $\|x_t - c\| \le r_{\text{out}}$ then $\langle x_t - x_0, e\rangle \ge \kappa\|x_t - x_0\|$ with e the unit vector to the centre, $L = \|c - x_0\|$ and $\kappa = \sqrt{L^2 - r_{\text{out}}^2}/L$ (expand $\|Z - Le\|^2 \le r_{\text{out}}^2$ and apply AM–GM). The thrust's coordinates are exchangeable and sign-symmetric, and so is any independent weighted sum of them; since at most $2/\kappa^2$ coordinates of any vector can each carry a $\kappa^2/2$ share of its squared norm, exchangeability alone gives $P(\langle Z, e\rangle \ge \kappa\|Z\|) \le 4/(n\kappa^2)$ and hence $r(n) \le 4h/(n\kappa^2)$ — an explicit bound with no concentration inequality in it. The truth is faster: measuring the cone event over 10,000 rollouts per dimension gives a clean exponential (log-linear $R^2 = 0.999$ versus log-log 0.952) at per-dimension factor 0.411, against the isotropic spherical-cap prediction $\sin\theta = r_{\text{out}}/L = 5/12 = 0.4167$ — agreement to 1.5%. So the collapse rate is the mode's *angular size from the start*, $r(n) \asymp (r_{\text{out}}/L)^n$, and nothing else: not its volume, not its thickness, not the horizon. The exponential law is moreover a *theorem* once the action interface is chosen well. If the thrust direction is uniform on S^{n-1}, the displacement is a sum of independent spherically symmetric vectors, hence spherically symmetric, so its direction is exactly uniform and the cone probability is exactly a spherical cap; with the elementary bound $P(\langle U, e\rangle \ge \kappa) \le \frac{1}{2}(1 - \kappa^2)^{(n-2)/2}$ this gives

$$r(n) \;\le\; \tfrac{h}{2}\,(r_{\text{out}}/L)^{n-2} \;=\; 40 \cdot 0.4167^{n-2} \quad \text{at the defaults,}$$

non-vacuous from $n \approx 7$ against $n \approx 400$ for the exchangeability bound, and verified at $n = 3 \ldots 8$ (simulated decay 0.431 against the predicted 0.4167).

The instrument's own thrust is cube-uniform and norm-capped, whose symmetry group is finite, so that argument does not apply to it — but a different structure does, and it gives the *sharp* rate. The norm cap depends only on the absolute values of the action, and the action's coordinates are independent and symmetric; hence conditionally on all absolute values the signs are i.i.d. Rademacher and the displacement coordinates are *independent*, each a Rademacher sum with deterministic weights. (Exactly so: flipping one sign changes its own coordinate and no other, bitwise.) Independence replaces symmetry: since the cone event forces $\sum_{i\ge 2} Z_i^2 \le \gamma^2 Z_1^2$ with $\gamma^2 = (1-\kappa^2)/\kappa^2$, a single Chernoff bound with matched exponent gives $P \le E[e^{\lambda\gamma^2 Z_1^2}]\prod_{i\ge 2} E[e^{-\lambda Z_i^2}]$, and optimizing λ in the Gaussian regime returns $\sqrt{en/(1+\gamma^2)}\,(1-\kappa^2)^{(n-1)/2}$: the cap rate exactly, losing only $\sqrt{n}$. Two ingredients close it. The first is Hoeffding sub-Gaussianity for Z_1. The second is that a

Rademacher sum is no more concentrated near zero than its Gaussian counterpart — which in its clean form is *false* (equal weights leave an atom at the origin), and whose correct form we prove by writing $E[e^{-\lambda Z^2}] = E_g \prod_s \cos(\sqrt{2\lambda}\, g\, c_s)$ and splitting on $|g|$: since $|\cos x| \le e^{-x^2/2}$ for $|x| \le 1.778$,

$$E[e^{-\lambda Z^2}] \;\le\; (1+2\lambda\sigma^2)^{-1/2} + 2\bar{\Phi}\Big(1.778\,\rho/\sqrt{2\lambda\sigma^2}\Big)\,, \qquad \rho = \sigma/\max_s |c_s|,$$

with no hypothesis on the weights for validity. The instrument's own conditional profile has ρ of median 4.0 and minimum 2.34 over $\approx$8000 samples, so the error costs $\approx 3\times 10^{-3}$: the per-dimension rate is 0.418 against the sharp 0.4167. Two things then matter about ρ. It is bounded below by 1 for free (Cauchy–Schwarz), and once λ is optimised *with* the error term included the resulting factor $(1+u)^{-1/2} + 2\bar{\Phi}(1.778/\sqrt{u})$ is below 1 for *every* $u > 0$, minimum 0.7783. That does not by itself give an exponential rate — a product of factors each merely below 1 need not be small, and a rate needs the $u_i = 2\lambda\sigma_i^2$ confined to a window — but it makes the window enormous ($q \le 0.89$ for σ_i^2 within a factor 5 of its mean either way), so only crude control of the per-coordinate scales is required. That control is unconditional. Since $\max(1, \|a_s\|^2) \in [1, n]$ deterministically, one has $\sigma_i^2 \ge \frac{1}{n}\sum_s w_s^2 a_{s,i}^2$ and $m \le \frac{1}{n}\sum_s w_s^2$, so the event that coordinate i is badly scaled is contained in one depending on *that coordinate's variables alone*. Two things follow: the exact Chernoff bound applies with the true moment generating function of a^2 (giving 9.7×10^{-4} where the range-based bound gives 0.31), and the bad-coordinate indicators are *independent*, so their count is binomial and Chernoff replaces Markov. The result is $P \le 2\cdot 0.9057^n$ — exponential, with no floor and nothing measured. Measurement then sharpens the constant: the observed spread $\tau \le 1.36$ gives 0.80^n, and $\rho \approx 4$ gives the true 0.4167^n. Details and machine checks are in the repository's theory notes (Lemmas C/E/F/G/ H/I, Theorems T5-C and T5-I, Proposition T5-T).

A methods note: the danger needs a competent planner, and "competent" is an interface property. An n-sweep whose vector-action MPC uses only the scalar planner's constant-candidate analogue measures play_cost ≈ 0 at every n. The cause is neither dimensional concentration nor the phantom's distance: that candidate set lacks the deterministic axial sequences $\pm e_i$, so the blind planner never drives straight at the shell. Direction-uniform random candidates do *not* fix it (pc = 0); adding the $2n$ axial candidates restores the 2D exploitation exactly (pc = 1.04, contact 1.0). The companion papers' clause that danger requires a *competent* planner is thus sharper than it looks: competence is a property of the action interface, and an incidental planner weakness can mask a fully exploitable model.

n	2	3	4	5	6
$r(n)$	0.0133	0.0033	0.0017	$0^\dagger$	$0^\dagger$
J truth-MPC	16.92	15.55	14.26	12.96	11.31
J random	0.50	0.33	0.04	0.01	0.07
$\mathrm{pc}_{\mathrm{blind}}$	1.023	1.013	0.994	0.991	0.996
blind contact	1.0	1.0	1.0	1.0	1.0

Table 7: The two axes are independent: rarity collapses with n (synthesis axis: mis-synthesis near-certain) while exploitation stays at full strength (play axis: the competent planner always reaches the shell). 600 rollouts per cell, `results/continuous_shellfield.json`; †censored — 0/600 observed, the 600 rollouts being independent samples within the cell, Wilson 95% upper 6.4×10^{-3}. $\mathrm{pc}_{\mathrm{blind}}$ with the axial-candidate planner (see the methods note in the text).

9.1 The non-separating control: obstruction is path-relative, gauge is separation-relative

One control remained to disentangle the thesis's two halves. A solid torus in $\mathbb{R}^3$ placed between start and lode (`TubeField3D`; thrust-vector action) does *not* separate space: an explicit around-path reaches the far side contact-free, so nothing is reach-null, there is *no exact gauge region*, and the crossing lemma has no separating surface to act on — certification-wise this mode is back in the companion papers' merely-rare world. Yet the danger dichotomy reproduces in full, governed by one knob that moves the torus's hole on or off the start–lode axis:

config	r	$\mathrm{pc}_{\mathrm{blind}}$	contact
hole on-axis (plan threads it)	0.0033	0.019	$0.00^{\dagger}$
tube on-axis (plan clips it)	0.0033	0.898	0.94

(†censored: 0/16 MPC episodes, Wilson upper 0.19.) Same rarity, same (trivial) topology, same instrument: only the position of the mode relative to the optimal path changes, and the exploitation swings from 0.02 to 0.90. The decomposition this completes: **danger is path-relative** (an on-path mode is exploited whether or not it separates anything), while **exact unfalsifiability is separation-relative** (only an enclosing boundary manufactures a reach-null gauge region). The ring, where the enclosing boundary and the blocked path coincide, conflates the two; the tube separates them.

The homological version of the query bound (Proposition 7), where the metric crossing has nothing to grip, turns out to be a *dichotomy*, and its unconditional form is false. Let K be the core circle, $g = \mathrm{dist}(\cdot, K)$ (1-Lipschitz), D the spanning disc, ρ_t the tube radius and $\Delta = 1.0$ the step bound (same as the ring's). If an interpolation point of a discrete path dips into the shrunken tube $\{g \le \rho_t - \Delta/2\}$, its nearer landing lies in the tube (clearance lemma); consequently a path with no landing in the tube can cross D only through the clearance sub-disc D_m of radius $R_c - (\rho_t - \Delta/2)$, and its linking number with K (equal to its signed D-crossing count; machine-checked against the Gauss integral) is realized entirely by hole-threadings. **Dichotomy:** a plan that links the core either queries the tube or threads D_m. **Refutation:** both linking classes contain landing-free — query-free — plans from start to lure, *even at the dangerous offset* (thread the displaced hole with clearance $2\rho_t$, or go around), so no positive query mass follows from topology alone: obstruction is path-relative as a *theorem*, and the measured 0.898 is the search's concentration near the straight corridor, not a geometric necessity. **What survives:** if every candidate's imagined path stays within ε of the straight segment, whose clearance to the core is $|o - R_c|$ exactly (offset knob o), then $|o - R_c| < \rho_t - \Delta/2 - \varepsilon$ forces $\mu_{\mathrm{query}} = 1$ — the registered offset 1.5 sits exactly at this boundary ($0.5 = \rho_t - \Delta/2$). Real trajectories, which never enter the tube (freeze), can link the core only through D_m: the hole is the only gate to the winding class under the true dynamics, and the offset knob's mechanism is visible in the real linking rate, $0.507 \to 0.283$ under an east-biased action arm. Proofs and machine checks are in the repository's theory notes (Lemmas X/Y, Theorem T8).

10 Related work

Code world models and verification-by-sampling. The CWM line synthesizes executable models searched by classical planners (Lehrach et al., 2026; Liang et al., 2023; Gao et al., 2023); the two companion papers (Aguilar Martín, 2026b,a) establish the danger law this paper extends and should be read first. Sampling-based acceptance is property-based testing (Claessen and Hughes,

2000) applied to dynamics; Proposition 1 is the exact statement of what such acceptance can pin down.

Model error in model-based RL. The pervasive-error framing (Janner et al., 2019; Lambert et al., 2020; Hafner et al., 2020; Nagabandi et al., 2018; Chua et al., 2018) is the backdrop the companion continuous paper argued against for hybrid systems; here the localized error is not merely rare but *structurally unreachable*, a regime that pervasive-error analyses do not model at all. The PAC model-learning line and the simulation lemma (Kearns and Singh, 2002; Strehl et al., 2009) bound value loss by model error *on visited states* — exactly the coverage the gauge region lacks by construction, which is why those bounds are silent here; safe learning-based control (Aswani et al., 2013; Berkenkamp et al., 2017) assumes the model's uncertainty is representable where it matters, the assumption the gauge region breaks.

Hybrid systems and reachability. Modes and guards are the bread of hybrid control (Paoletti et al., 2007; Bemporad and Morari, 1999); falsification tools (Annpureddy et al., 2011; Corso et al., 2021) search for violating trajectories and inherit exactly the reachability limit formalized here — a falsifier driven by any policy whose queries stay in $\mathcal{R}$ cannot exit $E(f)$. Reach-set computation and deductive verification (PHAVer (Frehse, 2005), SpaceEx (Frehse et al., 2011), Flow* (Chen et al., 2013), KeYmaera X (Fulton et al., 2015)) prove properties relative to a *given* hybrid model; our question is the dual — what a behavioral certificate can mean when the model itself was learned from reachability-limited evidence. Shielding (Alshiekh et al., 2018) assumes the very mode knowledge whose recoverability this paper measures.

Topological data analysis. Persistent homology and its stability (Edelsbrunner and Harer, 2010; Cohen-Steiner et al., 2007) supply the evidence sensor; homology-inference sample bounds (Niyogi et al., 2008) and the reach of a manifold (Federer, 1959) calibrate when that sensor can work, and persistence on dynamical and time-series data (Perea and Harer, 2015; Khasawneh and Munch, 2016) is the nearest use of TDA as a component of a running system. Section 8's contribution is not a new TDA method but a measured failure law for TDA *as a component of a certification loop*: the sensor's resolution propagates into the gate-certified artifact.

11 Limitations and scope

One instrument family, mostly round and separating. The core results live on round shells with freeze semantics, where the crossing lemma is metric and needs no homology (Section 3's honesty remark); the square ring ablates roundness and the solid torus now carries the non-separating case's linking dichotomy (Section 9.1). Non-round separators in general position (where Jordan–Brouwer is needed to define "inside") and moving boundaries remain the stated program, not results; the paper's claims are about what these instruments already suffice to separate.

Synthesis cells are modest; one contrast is underpowered. 20–30 seeds per cell across two GPT-5.x sizes; the cross-family evidence is spot-checks (three seeds per cell for Qwen and the Claude relay), which fix mechanism, not rates. The within-gap H2 contrast at $\gamma = 1.8$ remains directional-only even after doubling the cell to $n = 60$ (3/27 versus 2/33); the cross-gap crossover carries H2. The Qwen D cell is missing (provider credits), so the D-cell family contrast rests on GPT (1/20) versus Claude (3/3).

Two candidate mechanisms for the tail, both false. The heavy tail of the play-cost decomposition is not a freeze transient: the freeze counts of both continuations are 0/556, since after a dirty step both follow the truth planner, which knows the mode. Nor is it route commitment — the wrong action sending the continuation the other way around the annulus, so that the value function jumps across the cut locus. That would tie the planner-side analysis to this paper's own topology, but the correlation runs backwards: tail events split the two routes *less* often than the bulk (0.190 versus 0.299), and same-side pairs carry the larger advantages. What the tail is instead is a *delay* cost. Regressing the advantage on the difference in time the two continuations spend inside the phantom basin gives slope 0.94 against the reward amplitude 1.0 — a known constant, not a fitted one — with intercept ≈ 0 and $R^2 = 0.93$, and every tail event has a nonzero dwell difference against 41% of the bulk. One wrong action costs basin time. Two further measurements finish the picture. Dwell is sticky (neither planner ever leaves the basin once it arrives, 9/9), so the play-cost is exactly the blind planner's *lateness*: truth arrives at step 34–36, blind at 37–38. And that lateness is not a contact cost — in the facing configuration the blind trajectory never touches the mode at all. It is an *aim-point* cost: lacking a wall, the blind planner steers at the straight line to the lure, which a facing channel happens to thread. The degeneracy is correspondingly a knife edge. Rotating the channel by 0.4 radians turns "arrives four steps late, zero contacts" into "fifty contacts and never arrives". Since the executed path is contact-free, the two models agree along it, so the entire loss is candidate mis-ranking with no execution error.

That loss, however, cannot be bounded tightly from the model's own imagination, and this can be shown rather than conjectured. Every imagination-level quantity at a step — both models' returns on every candidate, and hence the model disagreement and μ_{query} — is a function of the state, the two models, and that step's candidate set. The realized per-step cost is a one-step deviation advantage of the truth policy, which depends in addition on the candidate sets the planner will draw at every later step, and those carry no information about the models. Pinning the imagination data exactly at one dirty step and varying only the future planner seeds moves the realized cost over $[-1.71, +0.60]$, a spread an order of magnitude larger than its mean over all dirty steps. So the vacuity of the companion paper's play_cost $\leq \mu_{\text{query}}(E)$ here is not an artifact to be sharpened away: a tight bound must condition on the planner's randomness as well, making play-cost a property of the model–planner pair rather than of the model alone — the same lesson the action-interface note draws from the other direction.

Conditioning that way does yield a bound. Averaging the truth policy's value over the planner's draws gives, exactly, $E[A_t \mid s_t, C_t] = \Delta r + \bar{W}(f(s_t, \tau)) - \bar{W}(f(s_t, b))$, a difference of one function at two states separated by a single action choice — and Proposition 4's landing law caps that separation with no further assumptions, at $2\,\text{gain}\,dt^2 = 0.06$ in position and $2\,\text{gain}\,dt = 0.6$ in velocity (measured maxima 0.059 and 0.590: the caps are tight). Everything then rests on how smooth the *seed-averaged* value is, which unlike the MPC policy itself is smoothed by the averaging. At a hundred CRN-paired seeds the smallest perturbation gives -0.015 ± 0.078, statistically zero, and the largest $+0.462 \pm 0.078$: a ratio of 0.78 ± 0.13, consistent with Lipschitz scaling. Combining yields pc $\lesssim 0.18$ against a measured 0.02–0.10 — but with that constant *measured*. Substituting instead the constant this argument proves (the drag cap times a per-step reward Lipschitz bound) gives 48, and the value it yields under the competence hypothesis of Proposition 7 gives 1.21; since play_cost ≤ 1 holds trivially, both are vacuous. The gap between the hypothesis chain and the measurement is worst-casing over the planner's action choices: every deviating step is charged the *maximum* velocity perturbation $2\,\text{gain}\,dt = 0.6$, and what the decomposition actually needs is the mean, since it bounds a sum. Writing the perturbation as $2\,\text{gain}\,dt\,|\sin((\phi_\tau - \phi_b)/2)|$, the mean is controlled by one inequality: that the angle between the two arg-maxes is no wider *on average* than between two independent uniform actions, whose $|\sin|$ has mean exactly $2/\pi$. Measured over

10,105 deviating steps across four gaps, two channel orientations and two planner budgets, that inequality separates the two orientations sharply. With the channel *facing*, every cell satisfies it (0.409–0.625 against $2/\pi = 0.637$), giving $\mathrm{pc} \leq 0.77$: non-vacuous, and with no fitted constant, since the measured input is an inequality between means rather than a number. With the channel *hidden* it fails in every cell, and instructively — the two arg-maxes are then near-antipodal ($|\sin| \approx 0.94$) and *every* step deviates, because a hidden channel makes the truth observationally a closed ring, so the truth planner turns west to the real lode while the blind planner charges east at the phantom. The bound is therefore a facing-channel statement, which is the configuration the aligned-channel degeneracy is about. What stays unproved is the inequality itself, a property of the model–planner pair — exactly what the negative result above says any tight bound must characterise.

The honest math ledger. The two γ-curve regularities (monotonicity of r_{int}, and that the wall never helps net entry) are theorems only *up to the funnel defect* (Corollary 2), which we bound by measurement (17× below the effect) rather than by proof. The routes to the unconditional statement are closed *with evidence*, not abandoned: the pathwise route by an explicit counterexample (seed 50543), the pointwise estimate by refutation in the data (91/91 CI-separated violations via a freeze-rescue effect). A third route closed the same way. Since freeze-rescue is caused by one modelling choice — a contact zeroes the velocity — a variant in which contact blocks the position but leaves the velocity updating freely would, if it made the ordering pathwise, isolate that choice as the sole obstruction. It does not: pathwise monotonicity fails in the variant as well, in 13 of 30,764 entering pairs (the unit here is the entering pair, since only those can falsify the ordering). The position block alone suffices to break it. What remains is an a-priori bound on one scalar, the funnel mass. A strictly positive answer already follows from Proposition 6, whose witness tube is freeze-free and therefore lower-bounds the *direct* component; but that bound carries the tube's factor ρ^h and is some ninety orders of magnitude below the measured value. The tube is the wrong instrument, and measurement says why: the direct-entry probability is a clean power law in the channel width (log-log slope 1.72 over $\gamma \in [0.05, 0.4]$, 200,000 rollouts per gap), i.e. a hitting probability rather than a tube probability. The plant explains it. In free flight the endpoint depends on each action with sensitivity gain $\pi\, dt^2(1-\beta^m)/(1-\beta)$ at lag m, rising to 3.14, so a *single* action sweeps the endpoint across some five units where one landing moves only gain $dt^2 = 0.03$; freeing two actions gives a Jacobian $K^2 W(T-s_1)W(T-s_2)|\sin(\phi_{s_1} - \phi_{s_2})|$ and a diffeomorphic image of area up to 39.5. So the exponential factor is an artifact of constraining all h actions when two suffice.

Asking for the probability that the *other* actions deliver the trajectory into position is, however, circular: a launch state is defined as one from which the remainder enters, so that probability is the entry probability again. A decomposition through the *ring-free* dynamics avoids this, because its factors cannot mention entry: $d(\gamma) = R \cdot \int_{\mathrm{channel}} \rho \cdot T(\gamma)$, with R the probability that a ring-free rollout ever reaches the band radius and ρ the density of its first-arrival angle — both independent of γ — and T the throughput of the channel. Measured, $R = 0.030$ and $\rho(\pi) = 1.02$ per radian (zero at $\pi/2$ and 0: the approach is one-sided, which is why a facing channel is the consequential case), and T has log-log slope 0.64, so the decomposition predicts $d \sim \gamma^{1.64}$ against the directly measured $\gamma^{1.72}$. Two independent estimates of T agree to 10%. The throughput has a kinematic account with a derived rather than fitted constant. The crossing takes $k \approx 9$ steps (the arrival radial speed is ≈1.5 against a band of thickness 1.5), while drag's time constant is $1/(\mathrm{drag} \cdot dt) \approx 33$ steps, so the arrival's tangential velocity persists ballistically through the crossing and survival inside an arc of half-width $r\gamma/2$ requires $|v_{\tan}| \leq r\gamma/(2k\, dt) \approx 2.2\gamma$. Since the density of $|v_{\tan}|$ at zero is bounded away from zero (measured ≈1.1, independent of γ), that probability is linear in γ and the throughput inherits it. The criterion tracks the measured non-freezing fraction across the range (0.25 versus 0.29, 0.50 versus 0.47, 0.93 versus 0.70, over-predicting at wide gaps where the ballistic approximation loosens), and the failure census confirms the corridor is what binds: freezing on the band accounts for 0.527 of channel arrivals at $\gamma = 0.2$ against 0.302 at 0.6, turning back out for 1–2%, and the horizon for the rest. Assembling the three factors gives $d(\gamma) \geq 0.0146\,\gamma^2$, which holds at every gap measured. Both one-dimensional densities the argument rests on admit lower bounds by the same elementary device: if a quantity is steerable by a single action whose law has a density, its own density is bounded below by $(2a_{\max})^{-1}$ divided by the steering sensitivity. For the tangential speed the designated step contributes gain $dt\,\sin(\phi - \theta)$ with ϕ uniform — the arcsine law, whose density exceeds $1/\pi$ on all of $(-1, 1)$ — giving $f(0) \geq 0.35$ against a measured 1.10. For the arrival angle, a single early action sweeps it over $\approx$1.1 radians, giving $\rho(\pi) \geq 0.94$ against a measured 1.02. The horizon term is bookkeeping and disappears once the deadline is built into the reach factor: counting only ring-free arrivals within $h - k$ steps leaves every arrival with the k steps the crossing needs, at a cost of a factor 1.4 (measured 0.0212 against 0.0302). The reach factor itself follows from isotropy, with no density estimate: the 2D heading action gives every thrust a uniform direction, so the displacement is a sum of independent isotropic vectors and is isotropic, whence its direction is uniform *and* independent of its magnitude. Reaching the band therefore needs the magnitude in $[L - r_{\text{out}}, L + r_{\text{out}}]$ and the direction within the ball's angular half-width, giving $R' \geq P(|Z_t| \in [7, 17]) \cdot \arcsin(5/17)/\pi = 0.019$ at $t = h - k$, against a measured 0.0212 — loose by 1.1. What remains measured is a single scalar, the radial law of an isotropic sum — and it is not a residue of this argument but the instrument's rarity itself. At $\gamma = 0$ the band spans the whole annulus, so a ring-free rollout reaches the band radius exactly when the same rollout would have contacted it: the two rates agree to five decimals (0.03020), and seed by seed without a single mismatch. The factor is therefore r, the quantity the danger law takes as its argument and this series measures per instrument. It also explains why no moment bound reaches it: the exact second moment of the Pearson walk puts the target at 1.3 times the rms displacement, so the event is in the tail — the mode's rarity and the failure of Chebyshev-type bounds are the same fact. Elsewhere the ledger has moved the other way, and we report both directions: the continuity modulus, the barcode sandwich, the non-separating query bound, and the play-cost decomposition are now theorems with explicit constants (some of them, like the refuted unconditional linking bound, negative results). Refutations are part of the record.

The sensor constant is ours; the mechanism is the claim. The resolution flip at $\gamma \approx 1.8$ belongs to one pre-registered detector at one density (Rips, dedup 0.05, cap 90, 3× median-NN). Other detectors move the constant; the propagated-resolution mechanism — artifacts tracking $\hat{\beta}_1$ rather than β_1 — is the claim, and it is detector-generic in the sense that any honest summary has *some* flip.

Normalization at wide γ. As the channel opens, the truth baseline itself changes (the phantom lode becomes reachable and J_{truth} jumps); `play_cost` is always reported with its own cell's baselines, and the danger conclusions rest on the paired contrast (facing versus hidden at equal γ), which shares the normalization issue symmetrically.

12 Conclusion

A sampling gate certifies f restricted to $\mathcal{R}$ — exactly that, and nothing else. On an enclosed mode this quotient stops being a formality and becomes the whole story: a wrong-topology artifact can be unfalsifiable by every gate and bitwise harmless at play; the same artifact, one knob later, is falsifiable and costly; the same omission, with the channel turned away from the planner, is at full danger again with the topology unchanged. Certifiability, correctness, and consequence are three different statements about where the model's errors sit relative to reach — never about the mode itself.

For practitioners the asymmetry is the message. A fence you cannot reach is a fence you cannot certify — and also one you cannot be harmed by while it stays unreachable; the danger lives exactly where the omission intersects a competent planner's optimal path, and it turns on over a knee as soon as a channel admits one step. Repair, when it is possible at all, needs two things that can both be measured *in advance*: evidence from the far side of the boundary (reachability of the interior), and an evidence sensor whose topological resolution at the operative sampling density can actually see the structure to be repaired. Where either is absent, the synthesis loop will do what ours did in every family we tested: write something plausible in the gauge region — and the gate will certify it.

Reproducibility

All experiments, result files, and analysis scripts are in the project repository:

- mechanism grid and γ-probes: `continuous_ring2d_mechanism.py`, `ring2d_rint_probe.py`;
- synthesis harness: `continuous_danger_synthesis.py`;
- agent-relay protocol: `continuous_claude_step.py`;
- registered open-ring driver: `continuous_ring2d_open_sweep.py`;
- aggregator: `ring2d_open_aggregate.py`;
- ShellField-n scripts and the behavioral audit (`ring2d_artifact_audit.py`, oracle self-tested);
- rarity calibration at every campaign configuration (`ring2d_rarity_sweep.py`, 30,000 roll-outs per configuration, configurations discovered from the committed campaigns rather than typed);
- held-out re-scoring of every artifact on disjoint gate and evaluation blocks: `heldout_gate_audit.py`, ring2d scope (block disjointness proved by set intersection at run time);
- the Lean formalization (`formal/`, library `Paper3Ring`; built by CI) — the deterministic results and the continuity modulus's measure steps, per the machine-checked remark in Section 3.

Every seed, checkpoint, and classified artifact is committed; long runs checkpoint per unit and resume exactly; paired seeds (common random numbers) are used for every play contrast. Each campaign JSON embeds its complete invocation (the harness's full argument namespace, under `params`) and the serving identity (`model` records the deployment name, created to equal the model version, so provenance is in the data rather than the lab notebook). Numbers in tables and prose are emitted by the named scripts from the committed JSONs, never hand-carried; a claims audit enforces this as a ratchet in CI.

References

Michał Adamaszek and Henry Adams. The vietoris–rips complexes of a circle. *Pacific Journal of Mathematics*, 290(1):1–40, 2017.

Javier Aguilar Martín. An omitted mode is a rare rule: The sampling-verification danger law in continuous code world models, 2026a. Companion paper (paper 2). arXiv:2608.17956, https://arxiv.org/abs/2608.17956; repository: https://github.com/JaviMaligno/code-world-models.

Javier Aguilar Martín. When a verified world model still loses: Play-adequacy vs prediction-accuracy in LLM-synthesized code world models, 2026b. Companion paper (paper 1). arXiv:2607.14169, https://arxiv.org/abs/2607.14169; repository: https://github.com/JaviMaligno/code-world-models.

Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, pages 2669–2678, 2018.

Yashwanth Annpureddy, Che Liu, Georgios Fainekos, and Sriram Sankaranarayanan. S-TaLiRo: A tool for temporal logic falsification for hybrid systems. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, volume 6605 of *Lecture Notes in Computer Science*, pages 254–257. Springer, 2011.

Anil Aswani, Humberto Gonzalez, S. Shankar Sastry, and Claire Tomlin. Provably safe and robust learning-based model predictive control. *Automatica*, 49(5):1216–1226, 2013.

Alberto Bemporad and Manfred Morari. Control of systems integrating logic, dynamics, and constraints. *Automatica*, 35(3):407–427, 1999.

Felix Berkenkamp, Matteo Turchetta, Angela P. Schoellig, and Andreas Krause. Safe model-based reinforcement learning with stability guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Xin Chen, Erika Ábrahám, and Sriram Sankaranarayanan. Flow*: An analyzer for non-linear hybrid systems. In *Computer Aided Verification (CAV)*, pages 258–263, 2013.

Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Koen Claessen and John Hughes. QuickCheck: A lightweight tool for random testing of Haskell programs. In *Proceedings of the Fifth ACM SIGPLAN International Conference on Functional Programming (ICFP)*, pages 268–279, 2000.

David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

Anthony Corso, Robert J. Moss, Mark Koren, Ritchie Lee, and Mykel J. Kochenderfer. A survey of algorithms for black-box safety validation of cyber-physical systems. *Journal of Artificial Intelligence Research*, 72:377–428, 2021.

Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction.* American Mathematical Society, 2010.

Herbert Federer. Curvature measures. *Transactions of the American Mathematical Society*, 93(3): 418–491, 1959.

Goran Frehse. PHAVer: Algorithmic verification of hybrid systems past HyTech. In *Hybrid Systems: Computation and Control (HSCC)*, pages 258–273, 2005.

Goran Frehse, Colas Le Guernic, Alexandre Donzé, Scott Cotton, Rajarshi Ray, Olivier Lebeltel, Rodolfo Ripado, Antoine Girard, Thao Dang, and Oded Maler. SpaceEx: Scalable verification of hybrid systems. In *Computer Aided Verification (CAV)*, pages 379–395, 2011.

Nathan Fulton, Stefan Mitsch, Jan-David Quesel, Marcus Völp, and André Platzer. KeYmaera X: An axiomatic tactical theorem prover for hybrid systems. In *International Conference on Automated Deduction (CADE)*, pages 527–538, 2015.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799, 2023.

Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.

Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Michael Kearns and Satinder Singh. Near-optimal reinforcement learning in polynomial time. *Machine Learning*, 49:209–232, 2002.

Firas A. Khasawneh and Elizabeth Munch. Chatter detection in turning using persistent homology. *Mechanical Systems and Signal Processing*, 70–71:527–541, 2016.

Nathan O. Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC)*, volume 120 of *Proceedings of Machine Learning Research*, pages 761–770, 2020. arXiv:2002.04523.

Wolfgang Lehrach, Daniel Hennes, Miguel Lazaro-Gredilla, Xinghua Lou, Carter Wendelken, Zun Li, Antoine Dedieu, Jordi Grau-Moya, Marc Lanctot, Atil Iscen, John Schultz, Marcus Chiam, Ian Gemp, Piotr Zielinski, Satinder Singh, and Kevin P. Murphy. Code World Models for general game playing. In *International Conference on Learning Representations (ICLR)*, 2026. Poster; OpenReview id 1UoB7IWiku.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 9493–9500, 2023.

Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, and Sergey Levine. Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

Partha Niyogi, Stephen Smale, and Shmuel Weinberger. Finding the homology of submanifolds with high confidence from random samples. *Discrete & Computational Geometry*, 39(1):419–441, 2008.

Simone Paoletti, Aleksandar Lj. Juloski, Giancarlo Ferrari-Trecate, and René Vidal. Identification of hybrid systems: A tutorial. *European Journal of Control*, 13(2-3):242–260, 2007.

Jose A. Perea and John Harer. Sliding windows and persistence: An application of topological methods to signal analysis. *Foundations of Computational Mathematics*, 15:799–838, 2015.

Alexander L. Strehl, Lihong Li, and Michael L. Littman. Reinforcement learning in finite MDPs: PAC analysis. *Journal of Machine Learning Research*, 10:2413–2444, 2009.